



Cyber-physical systems with Human-in-the-Loop: A systematic review of socio-technical perspectives[☆]

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ABSTRACT

Understanding and designing Cyber-physical systems (CPS) with humans in the loop (HITL) is a basic cross-scientific research problem with large implications for industry. The current software engineering knowledge already explains how to include the humans in the operation of the machines in terms of interfaces, architectures, adaptive systems, and design methodologies for including the Human-in-the-Loop. This paper extends existing knowledge with a systematic review of socio-technical perspectives on CPS with HITL. The review was software engineering focused, as it searched the body of research on CPS with HITL, and only within that body, those papers that included socio-technical perspectives. The results indicated four main areas in the ST literature. Validating these insights by expert interviews with industry CPS experts showed some alignment and also fundamental differences between the socio-technical literature (ST literature) insights and the industry experts' viewpoints. The discussion identifies useful crossings between the ST literature and research into CPS with HITL adaption, and touch on the issues of non-alignments in industry practice. The conclusion is that the ST perspectives on the body of knowledge on CPS with HITL has much to offer researchers in terms of innovative ways to look at the HITL, but the literature needs further development before industry experts can effectively use it. Future research possibilities are outlined.

1. Introduction

In recent years, Cyber-physical systems (CPS) with humans in the loop (HITL) (Nunes et al., 2018; Jirgl et al., 2018; Samad, 2023) have garnered significant attention across various domains, marking a transformative step in integrating computational processes with physical operations. These systems, which include critical infrastructures such as built environments (Bengler et al., 2024), transportation networks (Gil et al., 2019), and manufacturing processes (Emmanouilidis et al., 2019), leverage the synergy between human intelligence and automated processes to enhance operational efficiency and reliability. However, as technology progresses, the integration of human factors within CPS has become increasingly complex, necessitating sophisticated interactions between humans and machines. This integration is crucial not only for the operational success of these systems but also for their adaptability and resilience in dynamic environments.

Thus, despite the advancements in CPS, a fundamental challenge persists: the diverse ways of integrating human dimensions and their conceptualization. Current approaches often prioritize technological aspects without adequately considering the human operators' cognitive

and ergonomic needs. This oversight can lead to technically sound systems that fail to optimize human-machine interaction (HCI), resulting in reduced system effectiveness and potential safety hazards. Moreover, the lack of deep integration of human factors restricts the ability of CPS to adapt to new or unforeseen conditions in which human insight is invaluable. The literature reveals a significant gap in the systematic inclusion of human-centered design and development principles in the development of CPS. While there are methodologies to incorporate human factors, there is a scarcity of comprehensive frameworks that fully integrate socio-technical perspectives that consider the complex interplay between technology, individuals, tasks, and organizational structure (Leavitt, 1965). This gap suggests a need for a more nuanced approach that not only addresses the ergonomic and cognitive aspects of human interaction but also considers the broader socio-technical implications of these integrations. This paper contributes to the field by providing a systematic review of the current state of human factors integration in CPS and proposing a new approach that incorporates human-computer interaction principles. By focusing on socio-technical perspectives, this approach work seeks to bridge

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the existing gap by offering a comprehensive approach that enhances human-machine collaboration. The approach not only aims to improve system design and functionality but also ensures that the systems are adaptable, resilient, and capable of evolving with changing human and environmental conditions. This study identifies current characteristics, challenges, publication trends, and research gaps concerning CPS with HITL. The review is guided by four perspectives (Wohlin et al., 2015):

Purpose: provide visions and guidelines to consider CPS with HITL

Issue: identify, classify, and analyze four possible and necessary socio-technical perspectives on CPS with HITL

Object: Based on the existing CPS with HITL literature and interviews with industry experts, achieves a rich and extended CPS with HITL terminology

Viewpoint: For the software engineering research and industry communities, this paper offers a clear view of the HITL potential for CPS research and industry use.

The significant contributions of this paper are:

- Addressing an up-to-date state-of-the-art classification of CPS with HITL, focusing on gaps in software engineering addressed by socio-technical methods (see 'Results' chapter);
- Validating the findings with industry expert interviews (See 'Interviews with industry' chapter)
- Classifying CPS with HITL based on environment, infrastructure, coordination, and decision-making characteristics (See 'Discussion' chapter);

This study's address both research and industry communities interested in improving their knowledge and selecting suitable methods to design and develop their CPS with socio-technical considerations.

This study has been carried out according to systematic review guidelines provided in Kitchenham and Brereton (2013), Kitchenham and Charters (2007), Franzago et al. (2016, 2017). The following sections first present an overview of software engineering perspectives on CPS with HITL, then explain the method, including a selection of papers and taxonomy development. The main results are presented in four sub-chapters, followed by a general discussion, lessons learned, and a conclusion. Details on the extraction taxonomy is given in the accompanying data, see Clemmensen (2025).

Importantly, this paper focuses on the CPS with HITL literature and the socio-technical perspectives within that literature. There is thus a much larger literature from social science and humanities that does not directly concern CPS with HITL, but potentially could be relevant for understanding the HITL. As seen in our search string presented in the method section, we have excluded that literature in our search and selection of papers for the review, in order to focus on CPS with HITL research.

2. Background on cyber-physical systems

Cyber-physical systems (CPS) are sophisticated integrations of computational algorithms and physical components, facilitating seamless interaction between the digital and physical worlds (Greer et al., 2019; Baheti and Gill, 2011). CPS comprises interconnected networks of sensors, actuators, and computational elements, enabling real-time monitoring, control, and decision-making processes (Moghaddam et al., 2022b). CPS often overlooks critical human aspects (Wang et al., 2022) integral to their successful integration and operation in human-centered design and development and use, which considers human factors, ergonomics, user experiences in system development, safety, and comfort. Humans interact extensively with CPS, yet insufficient attention is given to their cognitive and behavioral requirements (Moghaddam et al., 2022a), leading to usability issues, decreased efficiency, and potential safety issues. CPS may also neglect socio-technical considerations (El-Haouzi et al., 2021), such as the impact on societal structures, human behavior, and ethical implications. For instance, in autonomous

transportation systems, the ethical dilemmas surrounding decision-making algorithms in critical situations remain largely unaddressed, posing significant societal acceptance and trust challenges (Reine et al., 2021). Considering such deficiencies, bridging the gap between CPS and human aspects requires a multidisciplinary approach called Cyber-physical systems with Human-in-the-Loop (CPS with HITL). The efficiency of the current CPS with HITL suffers from the following issues:

- **Incoherent Human-Centric Design and Development:** CPS are often developed with a primary focus on technical functionality and efficiency, with a brief representation of human needs, capabilities, and limitations, especially in the broader organizational perspective. This lack of a human-centric approach can lead to usability issues, cognitive overload, and reduced operator performance, ultimately impacting system effectiveness and safety.
- **Non-comprehensive Consideration of Human Factors:** Human factors, encompassing cognitive, behavioral, and ergonomic aspects, are essential for ensuring effective human-CPS interaction. However, CPS often fails to adequately account for these factors, resulting in suboptimal quality of experiences and potentially compromising system performance. Factors such as workload, situational awareness, decision-making processes, and user interfaces are crucial considerations that warrant greater attention in CPS design and operation.
- **Limited Incorporation of Socio-Technical Perspectives:** CPS with HITL are complex socio-technical ecosystems, involving interactions between technology, organizations, tasks, and human users. However, current CPS design and adaptation processes often lack a holistic socio-technical perspective, leading to challenges in addressing organizational dynamics, social impacts, and ethical considerations. Integrating socio-technical insights into CPS development can enhance system resilience, organizational readiness, and societal acceptance.
- **Insufficient User Involvement:** User involvement throughout the CPS lifecycle, including requirements elicitation, design iteration, and testing phases, is crucial for developing user-friendly and effective systems. Yet, many CPS projects fail to adequately engage end-users, resulting in mismatches between system capabilities and user needs. Incorporating user feedback, usability testing, and participatory design approaches can bridge this gap and ensure that CPS meets users' expectations and requirements.
- **Ethical and Societal Implications:** CPS decisions and actions can have significant ethical, legal, and societal implications, yet these considerations are often sidelined in favor of technical functionality. Issues such as algorithmic bias, privacy concerns, and societal impacts require careful attention and deliberation throughout the CPS lifecycle to ensure responsible and ethically sound deployment.
- **Lack of Adaptive Software and Feedback Mechanisms:** CPS software interfaces often lack adaptability and fail to react in real-time and provide real-time feedback to users, hindering their ability to effectively interact with the system. Adaptive interfaces, context-aware feedback mechanisms, and intuitive user interfaces are essential for facilitating seamless human-CPS interaction, particularly in dynamic and uncertain environments.

As the last issue points out, in CPS with HITL, adaptation entails intricate interactions between computational entities and human stakeholders (operators, programmers, architects, managers, etc.), necessitating a comprehensive understanding of their dependencies, links, interactions, and uncertainties. These uncertainties arise from various sources, including environmental dynamics, infrastructural changes, coordination complexities, and decision-making processes (Moghaddam et al., 2022b).

The **environmental** aspect encompasses the real-world context in which CPS operates, comprising both physical and virtual elements within or even beyond the system's direct control. These contextual dynamics significantly influence adaptation requirements and necessitate a nuanced understanding of the environment's contextual dimensions. Notably, humans are integral components of this environment, interacting with CPS, thereby introducing additional layers of complexity.

Infrastructure adaptation in CPS with HITL involves the reconfiguration of hardware and software components to accommodate changing operational needs. This entails the dynamic restructuring of sensor networks, computation resources, and actuators, alongside considerations of quality attributes such as performance, scalability, and security. While advancements in edge computing enhance CPS capabilities, challenges persist in ensuring seamless coordination and interoperability across heterogeneous components.

Coordination in CPS with HITL involves orchestrating interactions between dynamic infrastructures and evolving environments, including humans. Embedded distributed systems play a significant role in facilitating this coordination, leveraging sensors and actuators to influence the environment while adapting to contextual changes. Effective coordination strategies mitigate risks associated with system failures and resource constraints, ensuring operational and interaction resilience.

Decision-making processes in CPS with HITL depend on timely responses to uncertainties, necessitating proactive or reactive adaptation strategies. Decision methods, including model-based, rule-based, data-driven, optimization-based, and program-based approaches, guide adaptation processes based on system characteristics and goals. Proactive adaptation, enabled by (sophisticated) data-driven techniques, is favored for its ability to preemptively address quality degradation while respecting humans' quality of experience.

While existing literature often focuses on technical aspects of CPS adaptation, recent attention has shifted towards socio-technical perspectives such as human-centered approaches e.g., Lou et al. (2024), Liu et al. (2023), Lesch et al. (2023). Socio-technical systems involve the intertwined interactions between technology, organization, tasks, and human users, emphasizing the importance of human-centered design and development and organizational capacity enhancement. Integrating socio-technical insights into CPS with HITL can enhance system performance, user experience, and organizational resilience.

3. Method

As discussed below, the overall process can be divided into four main phases of search, selection, data extraction, and data synthesis. To achieve the research goal, we arranged for a set of questions along with their rationale and a definition of the key concept of the review.

3.1. Socio-technical concept and research questions

Socio-technical systems involve the interaction of hard systems (e.g., machines) and humans in ways that either cannot be separated or are thought to be inappropriate to separate (Law, 2009). The socio-technical concept emerged from the post-WWII, UK-based Tavistock institute's new approach to facilitating social change for groups of people, by system design. Many of the institute's staff had experiences from war time services of personnel selection, resettlement, and psychological warfare during WWII. Many of the staff were interested in individual psychological issues, since many of the staff were psychiatrists who believed in the power of deep understanding of humans, and because they saw human wellbeing as an important goal of industry practice. These experiences were the background on which the "action research" of socio-technical approaches developed, and why empirical analysis and theory development became embedded in organizational and individual change (Mumford, 2006). Later socio-technical system design approaches were build directly on the

original Tavistock institute's approach, with its focus on innovative and balanced relations between technology, organization, tasks, and IT users. Today, socio-technical system design is more relaxed theoretically and less guided by rigorous use of system theory and approaches that view technology and technology development as mostly technical issues (Bjørn-Andersen and Clemmensen, 2017). On the other hand, socio-technical design is more human-centered and less participatory in its ambitions compared to approaches often associated with Scandinavian system design (Bannon et al., 2018). Socio-technical system design in its many varieties aims for organizational capacity enhancement and support managements' interest, while at the same time caring for employees' experiences with human-computer interactions at work in industry.

On this backdrop, we developed a set of questions that look at CPS with HITL from social science and humanities point of view and helped to answer the overall research question: What are the socio-technical perspectives on CPS with HITL in the literature?

- **RQ1.1.** What are the socio-technical capacities of CPS with HITL?
- **RQ1.2.** How could we define a human-centered HITL approach in CPS.
- **RQ1.3.** What human-computer interaction technologies could be used for CPS with HITL?
- **RQ1.4.** What are dependencies of human work and CPS with HITL?

In addition to answering the RQs, we validated the insights from the literature review by using interviews to investigate the practical and industrial dimensions of CPS with HITL. Here we focused on how the application of CPS with HITL could be evaluated in practice and what are the associated challenges. We explain the method and results of that in Section 5 below, but first we present the selection, analysis, and results of reviewing the literature.

3.2. Search strategy development

A good search strategy is expected to provide practical solutions to the following questions: *which, where, what, and when* (Zhang et al., 2011).

Which Approaches? The search strategy consists of two phases: (i) automatic search in scientific databases; and (ii) snowballing. The first step was performed using the search string (see below) based on keywords from the research questions and areas of study. The search strings are used to retrieve potential primary studies through web search engines provided by digital libraries. Snowballing refers to using the reference list of a paper (backward snowballing) or the citations to the paper (forward snowballing) to identify additional papers (Wohlin, 2014). The start set for the snowballing procedure is composed of the selected papers retrieved by the automatic search, namely the primary studies, which are selected by applying inclusion/exclusion criteria to the automatic search results. In any case, the inclusion/exclusion criteria will be applied to each paper. If a paper is considered to be included, snowballing will be applied iteratively, and the procedure ends when no new papers can be found. In our case, instead of snowballing, we applied a second search string to systematically focus on the social science part of the identified software engineering literature on CPS with HITL (see below).

Where to search? The electronic databases used to search for relevant literature were ACM, Compendex and Inspec, IEEE and Scopus. Specifically to include SSH-oriented relevant research AIS e-library was added. In an experience report based on an SLR conducted in the area of agile methods, Dyba et al. (2007) found that electronic databases relevant to software engineering research, such as ScienceDirect and Wiley Science, returned similar search results as ACM, Compendex and IEEE. Hence, ScienceDirect and Wiley Science databases were excluded in our review. In addition, Scopus was selected since it claims to be the largest abstract database for peer-reviewed papers. For the first

Table 1
Electronic data sources targeted with search strings.

Library	Website
IEEE Xplore Digital Library	https://ieeexplore.ieee.org
ACM Digital Library	https://dl.acm.org
Scopus	https://www.scopus.com
AIS e-library	https://aisel.aisnet.org/

three search engines we searched the string on the full text, and for the Scopus we searched on title/abstract/keyword to limit the duplicate and false results.

What to search? A suitable search string was the input to the electronic data sources identified in the previous section, matching with paper titles, abstracts, and keywords. Following some test executions and refinements, the search string was finalized, as shown below. We codified the string in a way to be best adapted to specific syntax and criteria of each selected electronic data source. Further, we combined all studies into a single dataset, after removal of impurities and duplicates.

Search String After some tinkering, we finalized a search string that captured CPS with HITL papers across the selected databases, Table 1, which covered software engineering and social science, and which covered HITL-related concepts:

("human* in the loop") AND (CPS OR "cyber-physical system*" OR "cyber-physical system*" OR IoT OR "Internet of Things" OR IoB OR "Internet of Behavior*")

AND

socio-technical OR "socio-technical" OR "human-computer interaction" OR "human engineering" OR "work design" OR "work environment" OR "design of work systems" OR ergonomics OR "change management" OR "human systems" OR "cyber-physical social systems"

When and what period to search? We did not consider publication year as a criterion for the search and selection steps. Thus, all studies coming from the selection steps, until December 2022, were included regardless of their publication time.

3.3. Selection strategy

3.3.1. Paper selection

The paper selection process included steps for search, cleaning, and verifying the data set, Fig. 1. The final set of papers included 60 papers.

3.3.2. Data extraction

In this step, we classified the knowledge from selected studies by analyzing their content, as well as the general classifications in software engineering domain using keywords. When all studies had been analyzed, all keywords and concepts were combined to identify the research's context, nature, and contribution. We then performed a qualitative clustering operation to identify a set of representative attributes regarding specific aspects of CPS with HITL. The resulting taxonomy is presented in the results section, see also (Clemmensen, 2025).

3.3.3. Data synthesis

This activity involved collecting and summarizing the data extracted from the selected papers with the goal of understanding, analyzing, and classifying current research on CPS with HITL dimensions. Data synthesis was structured in two phases: vertical analysis and horizontal analysis. In vertical analysis data were analyzed individually to track the trends and collect information of each study. In horizontal analysis the possible relations across different dimensions and facets

of the research was focused. Based on the literature review step, we designed a primary framework to design the interviews around its various dimensions and components.

3.4. Interview study method

Following the expert interview method (Bogner et al., 2009), we recruited CPS experts from a body of industry companies engaged in the overall research project (see acknowledgment), with the purpose of doing a systematic interview study to validate the insights from the literature review. The criteria for selecting experts were that they should be in a position to put their own interpretations of CPS with HITL into practice. The interviewers were the first and second authors together with the aim of ensuring both social science and software engineering knowledge enough to do the interviews. Details on the participants and procedure and the derived interview questions are given below in the interview data section.

4. Analysis of the literature

In this chapter, we answer the question: What are the socio-technical perspectives on CPS with HITL in the literature?, by presenting and discussing relevant concepts extracted from the included body of 60 selected papers. See the method section for clarification. Fig. 2 summarizes the themes and sub-themes resulting from the literature review.

4.1. Socio-technical interactions in CPS with HITL (answer to RQ1.1)

The papers in this section discuss the role of humans in Human-in-the-Loop (HITL) within Cyber-physical systems (CPS) socio-technical systems. The focus is on the integration and governance of human roles within CPS with HITL to improve performance and decision-making. The section outlines how different levels of interaction between social and technical subsystems create multiple human roles. Examples include users reacting to changes in CPS, varying trust levels in interactions, and human-machine collaborations. It also touches on human-algorithm interactions, where humans act as agents, and how algorithms can influence task allocation, and how humans make sense of algorithm interactions. The section introduces the concept of socio-technical co-agency where humans and IT systems interact as equals, and humans are considered 'holons' in a holonic architecture. Additionally, the section describes socio-technical CPS networks as complex interactions among diverse agents, emphasizing the need for governance. Finally, it explores the importance of joint cognitive systems for continuous human-machine interaction, highlighting the enhancement of machine learning through human contribution.

4.1.1. The role of the human in socio-technical interactions

Socio-technical interactions include the interactions between the social (HITL) and the technical system (CPS). Socio-technical interactions create not one but several different human roles in CPS with HITL. For example, humans are seen as users in broad situations when the CPS displays report changes in the user's situational context that the user may have to react upon (Shao et al., 2019). Alternatively the humans are seen as someone with different and dynamic degrees of trust in the interactions (Brauner et al., 2019). A third example on a human role is as a part in human-machine interactions between the cyber and physical world, and in the interaction between the different actors and intelligent components in physical world (Tuunanen et al., 2015). A key question for all socio-technical interactions is thus who interacts with the technical.

Human-algorithm interactions models CPS with HITL as interactions between algorithms and human operators who are limited to key decision-making interfaces, or between algorithms and social media users in small "filter bubbles" of reactions and likes (Valdez and Ziefle,

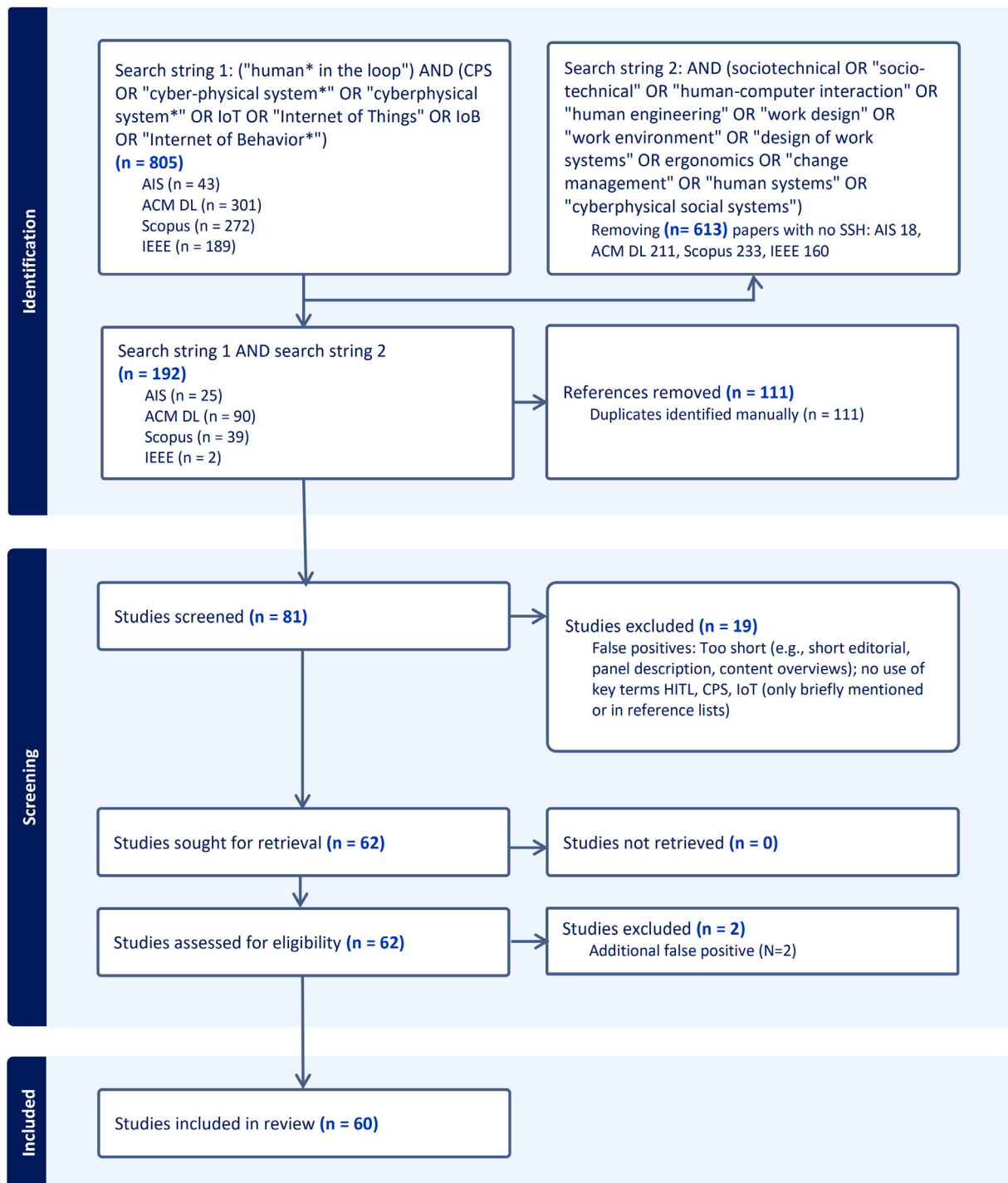


Fig. 1. The paper selection process.

2018), or simply how humans makes sense of CPS with HITL (Wolf and Blomberg, 2019). Such phenomena are understood by seeing human actors and users as agents. The algorithm can influence the task allocation by replacing humans or other algorithms, or by suggesting that others perform tasks, or even prohibit actors from doing specific important tasks (Martin, 2019). Human-algorithm interaction may not be only benevolent. For example, an algorithm requires provision from others of cleaned and ready data, which may be unethical (e.g., may expose workers to data that hurts to watch), or reversely, an algorithm by design or ownership hide data and data sources from users who would like to use those data (Martin, 2019). In concrete cases, human-algorithm

interaction may both provide useful work support and reconfigure work in negative ways, such as when helping in architectural design to identify outliers but also force upon users a backwards looking view with focus on historical experiences (Wolf and Blomberg, 2019).

Socio-technical co-agency occurs when humans and machines/IT systems work together on an equal footing. CPS is seen as a separate entity from the human in the HITL. While the interaction in the 'agent' perspective in the previous paragraph above is a closed loop between two entities, this alternative operates with interactions between separate actors. The relationship between them is named or at least called out as important (Fass and Gechter, 2015). The resulting system that includes

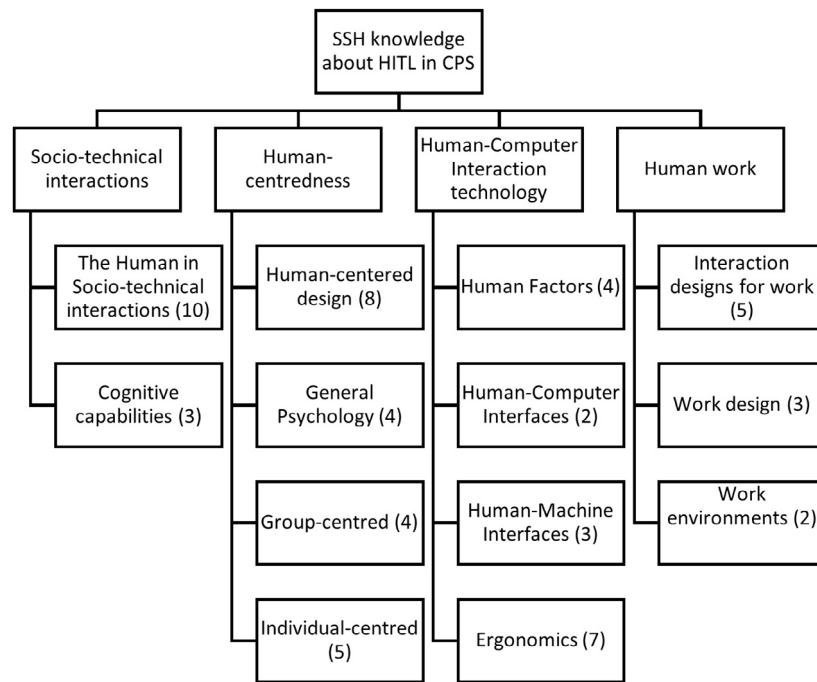


Fig. 2. Overview of Social Science and Humanities (SSH) knowledge about CPS with HITL in the 60 included papers. The number in each box is a count of papers focusing on the content of the box, see also (Clemmensen, 2025).

both humans and machines is a “socio-technical CPS” and can be modeled in a “holonic architecture” (Cimini et al., 2020). Humans are ‘holons’, that is, autonomous and cooperating conceptual entities that embed the interface needed to communicate with machines (Cimini et al., 2020).

Socio-technical CPS networks are socio-technical interactions between a variety of agents – humans, machines, productions, software, AI tools, CPS, etc. - built/grown with the purpose of completing physical or other, such as cognitive, tasks. They are thus very much networks of social interactions, and should be understood as such (Cimini et al., 2020). In this view, humans are social network nodes in multi-agent systems that need governance in the form of norms, policies, and preferences (Kampik et al., 2022). For example, humans and organizations are involved throughout design and use of urban sensor networks (Kleunen et al., 2021). Researchers and managers must address how normative governance concepts can be applied without sacrificing the benefit of autonomous agents, in particular in the networks in real-world applications at scale (Kampik et al., 2022). One way to do so may be through designing responsible and agile approaches to information use (Sadiq et al., 2022).

4.1.2. Cognitive capabilities in socio-technical systems

The cognitive capabilities in socio-technical systems concern the cognitive contribution of human activities towards the overall effectiveness of CPS with HITL. It is linked to the human ability to capture and act upon the industrial or other context of such interactions (Emmanouilidis et al., 2019). This view includes examples of having the human in the visual analytics loops (Emmanouilidis et al., 2019). In addition, techniques to address the cognitive capabilities at different levels in the socio-technical system (Gil et al., 2020) have been studied.

Having HITL can *amplify machine learning* (ML), for example by condition monitoring, and by extending ML capabilities by contributing to the (machine) learning process (Emmanouilidis et al., 2019). This adds human expert knowledge to data driven capabilities. It includes human verification of (other or own) human qualitative and vague interpretations of sensor data (Banerjee et al., 2020). *Enhancement of human capabilities* may also improve the (cognitive) capabilities of

the socio-technical CPS (Emmanouilidis et al., 2019). One example is reducing the need of local Cobot programming experts by creating a new economy of remote experts who act as HITL for the CPS by providing production instructions (Fitze et al., 2021). Another example is context-aware human machine collaboration that provides real-time assistance similar to an experienced human assistant during surgery (Fitze et al., 2021). The CPS-enhancements of human capabilities change human roles. In particular, handling *human-contributed micro-knowledge*, e.g., handling tagged maintenance meta-data, is important in situations with ill-defined problems, or in cases where there is a lack of rich data (Emmanouilidis et al., 2019). These phenomena are examples of effective methods for HITL in the machine learning loop for real application problems.

A *joint cognitive system* enhances the cognitive capabilities of the whole socio-technical CPS with HITL. Joint cognitive systems do not wait for direct actions on the system from a human agent in charge of decision-making, but acts in scenarios in which continuous interaction and data exchange between humans and machines occurs (Cimini et al., 2020). These systems are characterized by a high level of control of resources done technically, and the human supported by these as performer via a joint socio-technical solution (Emmanouilidis et al., 2019). For example, multiple CPS with HITL can be integrated by both vertical and horizontal integration of production systems, smart data analytics, autonomous agents, and support to humans by task-, context-, and user-adaptive interfaces (Brauner et al., 2022). Another example is how in-the-wild-testing of persuasive technologies can be enhanced by using testers that are hybrid entities consisting of a human agent and a technical device in varying combinations often leaning toward full automation due to the complexity of these situations (Brzezinski, 2018). Similarly, the fight against online fake can best be enhanced by having designs that help humans understand the rationale behind machine classifications and integrate human insights into machine learning, in forms of collaboration that depend on the context and audience (Te'eni and Ho, 2022).

In summary, socio-technical CPS with HITL is in the literature presented first in the category of *the role of humans in socio-technical interactions*, which includes *human-algorithm interactions*, *socio-technical co-agency*, and *socio-technical CPS networks*. Second, it is presented by the

category of *cognitive capabilities* in socio-technical systems, including *joint cognitive systems* and *enhancement of human capabilities*.

4.2. Human-centeredness of CPS with HITL (answer to RQ1.2)

The papers in this section emphasize the 'human-centered' enhancement of human sensory and actuation capabilities through CPS with HITL. Several papers attempt to define or demonstrate *human-centered design* by modeling humans as a system element. Other papers depart from psychological understandings of humans and their individual and group differences. They highlight the importance of understanding *general psychological phenomena* for reducing latencies and providing multi-modal capabilities, e.g., by establishing latency requirements for different senses. In addition, three forms of *group-centered CPS with HITL* are presented: human control models, human machine networks, and human team/group models, each with distinct roles and methodologies. Finally, *individual-centered CPS with HITL* addresses rational human behavior, system verification, and the need for reliable human behaviors, considering cognitive limitations and behavioral biases. Another group of individual-centered studies discusses reliable human behaviors, focusing on the study of human task performance and the use of simulators for training and assessment. Here, we also find papers that discuss digital human tactile skills, where wearable devices like HoloLens enable the transfer of expert skills to machines, enhancing task performance and allowing for the storage and reuse of these skills in similar situations. Overall, the section underscores the necessity of human-centered design and verification processes, agile methods, and the handling of uncertain interactions in CPS.

4.2.1. Human-centered design

Human-centered design of CPS with HITL models in various ways humans as system elements. Several papers see the human as a complex component that contributes errors to the CPS due to variations in human conditions. For example, variations in human health conditions can give errors, in which case the CPS, if the human is found to be unengaged with the task, disconnects the human from the on-going task to hinder further errors (Chui et al., 2018). Besides this treatment of humans as irrational decision makers that are best modeled as simple stochastic disturbances, it has been attempted to model humans as independent, intelligent agents who act within a control system to their own idiosyncratic benefits and who are influenced by incentive signals that can be used to control them (Bae et al., 2019). A closed-loop paradigm may generally model the human as a component involved in identification of semantic concepts and relationships, protocols for interaction, and iterative memorizing actions (Ma et al., 2018).

A slightly different modeling of the human as a component is the use of modeling average humans in iterative travel-demand simulations of physical infrastructure to provide feedback that real urban planners can use to do city planning (Feygin et al., 2020). Similarly, there is a focus on modeling abstracted humans when modeling user profiles, such as privacy profiles based on data sharing preferences, for enforcing compliance and ensuring user consent in software use scenarios (Kotut and McCrickard, 2022). Finally, human-centered computing includes modeling of typical human behavior is central to many applications of machine learning for CPS with HITL, for example, machine learning to improve smart cameras abilities to answer questions (Guo et al., 2018).

On a higher level, CPS modeling of humans in complex organizations can also be done by modeling workflows and other activities using motion sensors to enable management (scheduling) of rooms, with for example management of operation rooms in clinical settings (Joerger et al., 2018). Even higher level modeling of technology, organizations and people in multi-component networks are important to function construction and discovery of CPS with HITL, such as when framing the overall problem of designing systems for human missions to Mars (Boy, 2018).

4.2.2. General psychological phenomena of CPS with HITL

The *sensing and actuation role of human users* in CPS with HITL may be enhanced by the systems due to their physicality and ability to support physical capabilities of humans (Nunes et al., 2015). This differentiates the capabilities of CPS with HITL from cognitive technologies that can upgrade human communication skills and inference capacities by augmenting human capabilities in processing and sharing information (Cimini et al., 2020). *Human needs for sensing and actuating* are thus important for CPS with HITL (Fitze et al., 2021). Earlier promises of IT fulfilling human needs for information, entertainment, and social connectedness are now being extended by IoT and CPS to fulfill human needs for sensing and actuating (Fitze et al., 2021). However, to understand human needs for sensing and actuating, that is, for multisensory perception and action, requires considering individual human operator differences, including different expertise levels and age-related differences that are important to multisensory integration and delay requirements (Fitze et al., 2021).

Human multisensory perception and action are shaped by human-sense specific latencies in CPS with HITL. For example, reducing the interaction delay between the device and the user are important for in particular multi-device interactions (Hasan et al., 2019). Current research focuses on each of the human senses individually, and have so far established latency requirements from around 10 ms for video over about 3 ms for audio to only 1 ms for haptic, so the challenge is to both reduce the latencies and provide multi-modal capabilities (Fitze et al., 2021). For example, in intelligent lighting applications, human perception is important, but making predictions in a perceptual context involves several nonlinear, non-deterministic factors. These include psychophysics (with lots of noise in measurements), psychology (biases in perception of reality), cost/lifestyle (depend on the persons economy). Thus, new performance metrics that suit perceptual ML are required (Gopalakrishna et al., 2017). A somewhat related case is that of active vision with HITL, where tactile, sound, and verbal feedback combines in multi-modal interfaces to shape long-term co-adaptation of a human-smartphone system (Gallina et al., 2015). Thus, general models of human performance can be used as components in co-adaptation that allows systems to make inferences about changes in the users' cognitive and motor skills (Gallina et al., 2015).

4.2.3. Group-centered CPS with HITL

Group-centered CPS with HITL appears to come in three forms: *human control models*, *human machine networks*, and *human team/group models*. The idea in human control models for CPS with HITL is to consider the human as a crucial part of the whole system, which is very different from and in contrast to autonomous CPSs that do not have any human components (Jirgl et al., 2019). The human control model in CPS with HITL builds on human reliability assessments assuming that the human is important to deal with unexpected situations and tasks, but also suffers from fatigue and makes human errors (Jirgl et al., 2019). In addition, the human control takes place by the human interacting with other parts of the CPS through human-machine interfaces (Jirgl et al., 2019). Thus, the human control models related to CPS with HITL need to be complex enough to predict task-specific human behavior on different levels of sensing and cognition, and also able to adapt to changes over time in human-machine collaboration.

The idea in *human machine networking* for CPS with HITL is to enable the machines to dynamically augment the network operations to the benefit of humans, beyond simply relaying information, and towards dealing with tight latency requirements for multisensory and high bandwidth requirement scenarios in adjustable networks (Fitze et al., 2021). One example is swarm collaboration systems for smart manufacturing that are design to create human-machine social networks for harmonious interaction between human beings and computer systems in scenarios with multiple dynamic IoT events (Yang et al., 2018). Another example of this practice is leveraging AI and machine learning techniques for audio quality improvements (e.g., Fitze et al. (2021)),

as well as video and screen sharing optimization in communication and collaboration platforms. This kind of human machine networks enables novel human experiences, e.g., by allowing humans to interrupt each other while talking during online calling and meetings, in accordance with basic human needs for sensing and actuating.

Human team models comprise a third approach to group-centered CPS with HITL. The idea is self-integration, where subsystems (e.g., human and software/hardware agents) are equipped with capabilities to continuously assess and maintain their own integration status within the overall system composition (Botev, 2020). Thus, the actions and reactions of the humans play a central role towards self-awareness, self-adaptation and self-integration of the various artificial agents in the system, while the humans, conversely, need data and visualizations to become aware of interactions between artificial agents and the way these are communicated to the other system constituents (Botev, 2020). Furthermore, to co-adapt models of humans for CPS with HITL, the HITL component needs to actively ensure the well-being of the human operators in-the-loop, who again may come from heterogeneous populations with different issues of access (Fitze et al., 2021). Hence fundamentally different design methodologies are used, and a purpose-driven combination of different modeling approaches (e.g., modeling from a technical-rational perspective and modeling from a human actor perspective) may be the best way forwards for the design of human team and group models for CPS with HITL (Botev, 2020). The differences between these approaches are subtle and encompass issues such as the requirements for CPS with HITL to adapt to changes over time in human-machine collaboration (human control models), enable basic sensing and actuating human experiences (human machine networks), and actively ensure the well-being of the human operators in-the-loop (human team/group models).

4.2.4. Individual-centered CPS with HITL

Individual-centered CPS with HITL concerns issues of *rational human behavior*, *human-centered system-verification*, *reliable human behaviors* in HITL, and *digital human tactile skills* towards the physicality of the CPS. *Rational human behavior* is central for human decision making from a behavioral economics perspective, which points out that behavioral biases affect decision making and thus may hamper CPS with HITL by introducing sources of uncertainty (Protte et al., 2020). In particular, humans have bounded rationality in the sense that they are strategic thinkers who aim to maximize their benefits but are limited cognitively, hence prone to behavioral biases such as over-confidence and misperception of random sequences of events (Protte et al., 2020; Bae et al., 2019). The latter happens when the decision maker comes to believe that the sequence of events exhibit a pattern that should occur again; this may occur in closed-loop scenarios where the decision maker receives feedback about the success of its actions, so in that sense open-loop scenarios without such feedback information may be safer (Protte et al., 2020). Human rationality considerations include how to design for fairness with adaptive CPS with HITL that can address intra-human and inter-human variability by monitoring the human state and environment changes through IoT sensors; the idea is to ensure fairness for multiple humans sharing the same IoT application space such as cars and homes (Elmalaki, 2021).

The idea in having the *human at the center of verification* is that the human must be a persistent center for all design and verification processes of CPS with HITL, including those done post-implementation in everyday work situations with real sensor data at play. The human-centeredness concerns the design and verification of all physical and electro-mechanical embedded systems, as well as integrated high-level software executed on mobile devices, computing platforms, and cloud-based resources (Cuckov et al., 2017). This complexity can be handled by agile design and construction methods with feedback loops between consecutive development stages for immediate and rapid testing and re-design, and between non-consecutive stages for requirements checks, enforced through measurements and systematic compliance

tests (Cuckov et al., 2017). Involving humans introduces uncertain or chaotic interactions in CPS. Human interpretation of sensor data is often vague and of qualitative nature. There are different ways to deal with this. Similar but slightly different, it has been suggested to let users choose from trace-generated trigger-action rules to let individual users follow their own automation priorities (Zhang et al., 2020). So in contrast to either fully relying on users to provide input or fully relying of automated learning to infer automation models, a combination of these approaches may help effectively aligning the humans with the computer (Zhang et al., 2020). Another approach is to let sensor information can be formally approximated, verified, and identified by humans in graded terms such as low, medium, high, etc., that is, noise from different sources (i.e., sensors) in the overall system can be expressed “linguistically” rather than as quantities. Banerjee et al. (2020). Thus, an assessment of the CPS with HITL’s safety can be done and delivered with a human-interpretable metric (Banerjee et al., 2020). To do this may be of key importance for positioning the human at the center of verification of CPS with HITL.

Evaluating reliable human behaviors is another individual-centered issue in CPS with HITL. A general approach to human reliability is the attempt to model the probability of individual human error based on models of control feedback, coordination rules, and organization and cognition knowledge (Jirgl et al., 2018). Thus the principle of human reliability of CPS with HITL consists in studying human task performance within various operating scenarios and evaluating the reliability of the individual human behaviors with the purpose of failure and human error prevention (Jirgl et al., 2019). These evaluations are based on the assumption that human behavior is explainable as information processing in the brain. However, information about behavior related to the observed CPS with HITL task is often unavailable, and instead the assessment relies on predefined probabilistic values of parameters for tasks and operations provided in standards (Jirgl et al., 2019). An alternative to naturalistic observation of CPS with HITL is to use a suitable simulator equipped with data acquisition to analyze (and train) human behavior within different standard and critical scenarios. A simulator approach to data collection also allows qualitative assessment of the human behavior to describe more accurately the phenomena (Jirgl et al., 2019).

The idea of *digital human tactile skills* is related to the physicality of the CPS with HITL. With wearable devices (e.g., HoloLens) and their built-in tracking of human motion and mapping the sensed information to the human behavior and robot actions, experts can transfer skills from human to machines (e.g., how to close the water flow to repair a sink). In addition, the combination of sensors and actuators can be employed in the wearables, constituting part of these user interfaces, and control the HITL to enable expert task performance by novices following instructions and guidance from the machines. For example, specific gestures or motion paths can be generated and the human operator be notified when deviating from the planned path (Fitze et al., 2021). Furthermore, such digital human tactile skills can then be stored and used in new situations of the same type (Fitze et al., 2021).

Thus, individual-centered CPS with HITL offers both challenges of how to design for qualitative and vague human beliefs and interpretations (e.g., human-centered system-verification, rational and reliable human behaviors in HITL), and great promises of shareable digital human skills (e.g., digital human tactile skills towards the physicality of the CPS).

4.3. Human–computer interaction (HCI) technology for CPS with HITL (answer to RQ1.3)

The papers that are about HCI perspectives on CPS with HITL suggests that humans should be considered both service consumers and providers within CPS, contributing not only factual data but also judgments and opinions. It advocates for adaptive interfaces that co-evolve

with user skills and for empowering human operators with autonomy in decision-making processes, even in highly automated environments. This approach aims to harness human sensory capabilities and cognitive skills to enhance system performance and safety. The first group of papers in this section address the integration of human factors in system design, emphasizing the importance of human roles such as the actor, decision maker, and operator. Other papers highlight the evolution of human-machine interfaces from analogue to digital, stressing the need for interfaces that adapt to user context and minimize attention costs. Finally, a group of papers focus on ergonomics, emphasizing safety, skill evolution, and autonomy.

4.3.1. Human factor

Across HCI technologies, notions of the human include *human factor*, human actor, human decision maker, and human operator, depending on the context and work role of the human. The classic human factor notion considers humans as a system component just as or perhaps even more important than the software and hardware, but with a tendency to be the most unreliable component of any CPS with HITL (Cuckov et al., 2017). For example, human factors (e.g., human errors and driver medical conditions) are critical reasons for car crashes (Chen et al., 2018). In this view, human factors include training level, stress, or fatigue, and the range of ways in which humans can participate and the roles they can play when collaborating with autonomous CPS should be explored (Gil et al., 2019). Human factors may be seen even broader – and more positively – and include supervision, setting adjustment, knowledge and competences, situation diagnosis, and in general as those providing additional degrees of freedom to the CPS, as ‘flexibility drivers’ (Fantini et al., 2016). The human factor can include engineering, monitoring, controlling, and performing operational maintenance roles, with the human component being different from the other parts of the CPS in the sense of being biological (Fass and Gechter, 2015). Other notions of the human, however, consider the human as more than an unreliable factor. Instead, the human is seen as an actor; thus the human becomes central in the design of CPS with HITL when moving from the notion of information to one of sensing and actuating, and with human and machine operators seamlessly connected (Fitze et al., 2021). CPS with HITL builds on collective evidence from the technical and human system actors (Emmanouilidis et al., 2019). The human decision maker is a risk factor in CPS with HITL. This implies that it is necessary to know about human decision makers’ abilities, weaknesses, and potential biases in each task-domain or risk-context, in order to design support (e.g., visualizations) for human decision makers’ trust, (blind) compliance, and the restoration of trust after a failure (Brauner et al., 2019). Finally, the human operator in CPS with HITL is always a perceiver of the environment (Fitze et al., 2021), so designing for work contexts is of key importance.

4.3.2. Human–computer interface

While HCI in general is about interactions, the human–computer interface is a specific material component in the computer system in the same way as the digital, analogue, and cyber components (Cuckov et al., 2017). These components, including the human–computer interface, are connected within the system in complex ways (Cuckov et al., 2017). For example, to establish a human–computer interaction platform for safety testing of HITL for take-over situations in autonomous electrical vehicles driving, Logitech G29 Driving Force Steering Wheels and Pedals were used as human–computer interfaces to enable human control of vehicle steering angle, gear, gas pedal position and brake pedal position (Hong and Aparow, 2021). Human–computer interaction may thus take place using the human–computer interface.

4.3.3. Human-machine interface

Human-machine interfaces historically precede human–computer interfaces. Analogue gauges, buttons, levers, alarms, switches, etc. have

appeared in transport, manufacturing, and military domains for many decades (Singh and Mahmoud, 2019), and actually throughout human history. While the interfaces may always have been human-centered, today there is a strong need for human-machine interfaces that support changes in the user’s situational context and minimize human attention costs (Shao et al., 2019). This requires capture and analysis of what happens in the human-machine interface to detect human errors and support human performance in CPS with HITL in real time (Singh and Mahmoud, 2017). Related to this, it remains a specific problem how to replace or monitor existing legacy human-machine interfaces with all their human-sense-dependent analogue controls and displays and related expert human skills (Singh and Mahmoud, 2019). HCI has a variety of notions of interface history and a long tradition for ergonomic interfaces.

4.3.4. Ergonomics of CPS with HITL

Ergonomics traditionally covers most of human–computer interactions (the term is sometimes used synonymous with HCI), but it also covers physical ergonomics such as how workers interact with chairs, lifts objects, use machines, etc. For CPS with HITL it is important to look at all aspects of ergonomics, including *ergonomic methods* such as novel methods for machine and human sensing for safety, *adjustment of HITL over time* as human skills evolve, and the provision of *human autonomy*.

While traditional *ergonomic methods* of CPS with HITL aim to ensure the safety of humans are passive structures, newer methods provide sensing-driven safety, that is, considers machine and human sensing for safety. This includes what have been called neuroergonomics, that is, physiological measurements to understand an human behavior by recording physiological changes of CPS with HITL (Nuamah and Seong, 2017). For example safety around factory robots involve the use of cages, while newer Cobots promise to provide sense-driven solutions for human safety around robots (Dani et al., 2020). The assumption is that it is possible to infer human intentions from sensing information; challenges for this include ensuring quality of long term intention estimation (e.g., does the human behave in the same way today as yesterday?) and fusing information from multiple sensors including some that humans do not have such as Lidar and physiological signals (Dani et al., 2020), and then reliably use this information about human intentions to increase safety in real time Cobot scenarios, following the principles of human-centered design in the ISO standard on ergonomics of human–system interaction (Gil et al., 2020). Furthermore, in ergonomics humans can be viewed as not only service consumers consuming, e.g., safety, but also as service providers, who are sensed by sensors or directly answers sensing requests from the CPS (Domaszewicz and Parzych, 2022). The humans are not in charge within the CPS in these cases, but deliver ‘sensing’ services to the organization owning the CPS (Domaszewicz and Parzych, 2022). Among the different benefits of viewing humans as service providers in the CPS with HITL is that this view utilizes the vagueness of human sensing as a strength; thus human employees can offer not only facts, but also vague opinions and judgment (e.g., about safety in a broader sense such as the ergonomics of a chair, work overload, satisfaction, mobbing) (Domaszewicz and Parzych, 2022).

CPS with HITL can be *adjusted over time* by capturing changes in motor and cognitive abilities of the users over time, and continuously adjust the human-machine interface to the level of the users skills (Gallina et al., 2015). For example, the joy-stick operation of excavators, cranes, forklifts, electric-powered wheelchairs, and tele-manipulated robots can be adjusted over the years as users learn to use it, so that the human and the interface progressively co-adapt (Gallina et al., 2015). This raises questions about how smart human interfaces with micro-learning opportunities, easy programming, and presentation of a diversity of information generated by model-integrated AI, can enable task, context, and user-adaptive interactions (Brauner et al., 2022) – and continue to do this over time.

CPS with HITL provides a level of *human autonomy*, since humans in many cases not only perform tasks, but are continuously involved in all aspects of associated decision making (Minnetti et al., 2020). For example, in zero-defect, multistage industry 4.0 manufacturing where humans perform quality control tasks, they can be considered smart quality control stations that show real-time adaptive behaviors that help keep measurement uncertainty under control, including in situations with variations in process or product parameters. In the case of car manufacturing, for example, many assembly-line operations are done by the human operator that manually assembles the car parts and iteratively measures gap (tangent space between two adjacent surfaces) and flush (mismatch between two orthogonal surfaces), and then corrects the alignment until measurements fall within specifications (the importance of car body parts alignment are well known to car customers who see this as a quality of the assembly of their new car). However, the problem is how to enable the human operator working in an assembly line to take gap and flush measurements and use data locally for quality control and improving the assembly as well as store data on the factory database (Minnetti et al., 2020). Empowering human operators with autonomy, e.g., by providing a portable manually operated hand-held gap and flush measurement device that support the human actively engaged in quality and process control decisions, fits in the context of CPS with HITL (Minnetti et al., 2020). On the other hand, ergonomic paradigms for HITL such as ‘soft actuation’ indicates that neither full automation, nor forcing an explicitly active human role in the CPS, are always the best solutions (Domaszewicz et al., 2016) instead the use non-intrusive ‘hints’ from sensing systems can allow the human to choose when to be an active actuator in the CPS (Domaszewicz et al., 2016).

However, humans tend to be increasingly obedient to faulty systems with increasing task complexity (Brauner and Ziefle, 2019). For example, in CPS with HITL in a cybersecurity context, it turned out that the human operator of autonomous vehicles, would become more suspicious of malign intent behind alerts from a operator support system and would take longer time to investigate and respond in scenarios with cyberattacks and in more important scenarios (Gay et al., 2017). Therefore, understanding employee cognition of information from security tools as in influence on user attitudes towards security policy compliance may help find the best ways to encourage employees in small and medium sized business to comply (Adams and Liu, 2022). Thus the ergonomics of CPS with HITL touches on “soft” social science topics like suspicion and obedience.

4.4. Human work design for CPS with HITL (answer to RQ1.4)

The papers in this section examine the role of CPS with HITL within the context of human work. The section discusses the scarce literature on HITL and CPS in work-related fields, and presents findings on interaction designs, work design, and work environment. *Interaction designs for work* emphasize iterative testing and redesigning to achieve suitable interaction within smart work environments. *Work design* focuses on socio-technical approaches, integrating human and machine collaboration, and considering extreme work scenarios like Martian colonies. *Human work environment* highlights the importance of human control in automated systems for maintaining productivity and satisfaction, suggesting a need for innovative, safe, and trustworthy CPS with HITL. The overall aim is to create work systems that are adaptable, efficient, and enhance human-machine collaboration.

4.4.1. Interaction designs for work

Designing prototypes can be viewed as a tool for helping to improve CPS with HITL interaction designs for work, since it serves to iteratively test and redesign the prototype until a suitable interaction design is achieved (Gil et al., 2020). Interaction designs for CPS with HITL share the common goal of changing the way interaction is realized towards new forms of smart work environments in which the boundaries of

virtual and physical worlds dissolve and novel forms of interaction are applied (Magerkurth, 2012). In human work, CPS with HITL requires the interaction design to follow push and pull principles so that both sides can request resources (e.g., the computer can employ an active learning approach and signal the need for additional labels as a data modification from the human, and the human can request information and operations from the computer) (Zschech et al., 2021). Moderating factors on the interaction design can include work task’s criticality, e.g., criticality of vision-based detection tasks in automotive, energy, manufacturing, agriculture, architecture work domains, the experience of the users (workers), and other factors (Zschech et al., 2021). For example, interaction designs of alarm systems for vulnerable groups, such as migrants, homeless, and home violence victims, require CPS with HITL to protect against attacks by gesture and voice recognition that can be activated either automatically or by the human pressing the alarm button (Silva et al., 2019). A similar need for an interaction design for work that aims to balance the HITL is obvious when it comes to designing the best moment to let smart speakers talk to the humans next to them; for example participants in their own homes accepted to be interrupted by smart speakers in 18 percent of the cases while doing paid work versus 85 percent while doing unpaid household work (Cha et al., 2020). CPS with HITL requires interaction designs that relates humans to the objects around them, and the wider infrastructure in which the CPS are embedded (Murray-Rust et al., 2022).

4.4.2. Work design for CPS with HITL

Work design for CPS with HITL that enable smart workplaces and smart work environments may concern technologies that support human work, collaborations between humans and machines, social interactions in complex systems, and more (Cimini et al., 2020). The human worker configuration of CPS with HITL is a part of this; the context of work design includes the type of production system in which work takes place, including task allocation between machines and humans, and the machines’ autonomy and the general level of automation in the work environment (Cimini et al., 2020). To achieve high quality work design, the *approach to the customers* regarding the design of the manufacturing process and human work design should be done in a socio-technical way, that is, designing the social and technical sub-systems together. For example, solutions for structuring image data in the fields of medicine, traffic and machinery production require pre-defined sets of images that first must be provided to enable crowd workers to perform the labeling (Schuir et al., 2021). The crowd work of labeling, crowd workers’ knowledge, organization of the crowd work, *norms, policies*, and suitable technical tools (e.g., a and augmented reality solution), must all be designed concurrently (Schuir et al., 2021).

The human work design technology itself may need to be further developed to design for a variety of novel work systems (Brauner et al., 2019; Cimini et al., 2020). To *innovate work design* for CPS with HITL, it pays to consider extreme scenarios. Imagine the work environment on future Mars or Moon colonies. A Martian workplace (Linder et al., 2022), may require a very different work design from current workplaces on Earth. The type of work task, isolation from other human societies, and extreme setting call for work design for adaptability, extreme asynchronous communication, CPS with HITL guided by an extremely small number of qualified people, and solutions for stress mitigation in highly distributed work contexts. In general, work design encompasses the design of organizational structure, employees/people, tasks, and CPS with HITL, as one entangled socio-technical work system b (Brauner et al., 2019). Interestingly, for extreme scenarios like the workplaces on future Martian colonies (Linder et al., 2022), the work systems to be designed may span two or more planets.

Table 2
Industry expert participants.

ID	Jobtitle	Expertise area	Expertise level (experience in industry)	Education level	organization
P1	Manager of research collaborations	Research and technology development and strategy	4 years	PhD in computer science (theoretical)	Global leader in the variable speed control of electric motor
P2	Head of incubation and partnerships	Product development	15 years	Business candidate in international management	Established innovative Cobot production company
P3	Project manager	Responsible for R and D projects	14 years	Master's degree in electrical engineering (programming)	Global leader in insulation products
P4	PhD fellow, specialist	Cobot design	3 years	Master's degree in SE	Established innovative Cobot production company

4.4.3. Work environment

Workplaces are increasingly found within *smart work environments* such as provided by building automation and connected services. Building automation and centralized control systems provide indoor comfort, e.g., HITL for CPS that provide climate management, light (Gopalakrishna et al., 2017), security registration, and other services, and conserve energy. However, limiting or removing employees' control over their work environment by automation with CPS results in dissatisfaction, and hence decreases productivity (Loengbudnark et al., 2022). Having CPS with HITL and hence human control is thus directly and positively influencing the productivity. The collaborative work environment between humans and robotic automation in building automation and other work environments may require innovations in safe, efficient, and trustworthy CPS with HITL (Dani et al., 2020).

5. Validation of literature review: Interviews with industry experts

In this section, we validate the literature review insights by applying structured expert interviewing with industry experts, on questions derived from the literature review results (Bogner et al., 2009).

5.1. Participants and procedure

The participants were four experts in research and development of cyber-physical systems in industry, Table 2. The interviews were individual, varied from 43 to 75 min, and were done by the first and second authors together. The interviews were done via MS teams and using their transcription, and analyzed with the qualitative data analysis software MAXQDA 2024. The analysis of the interviews was done as based on questions that we systematically derived from our literature review results Sections 4.1 and 4.4. This allowed us to compare directly the insights from the literature review with quotations from the interviewee participants in order to shed light on the usefulness and alignment of literature insights and industry experts viewpoints. All the interview questions that are presented in the following sections were preceded with "In your organization...".

5.2. Interviewees' response to general socio-technical issues related to CPS with HITL

This section presents the industry experts responses to the interview questions on socio-technical interactions, Table 3.

Table 3

Interview questions on socio-technical interactions, derived from Section 4.1.

Socio-technical interactions and the human role
Who are the humans in CPS with HITL? (Situational users, trusting humans, intelligent actors, other?)
What are the benefits and downsides of seeing human users as agents involved in human-algorithm interaction? (for decision making, social support (social media bubbles), task allocation, human access to data, other?)
In which ways are the humans on an equal footing with the CPS when they work together on a task? (humans and machines are equally involved in the HITL? Or other forms of co-agency, co-work?)
What are the norms, policies, and preferences for social networks/groups/teams that also applies to machine agents? (If none, how would you develop such normative approaches?)
How do you measure the (cognitive) performance of CPS with HITL? (which contexts can the human in the CPS with HITL capture and act upon?)
Cognitive capabilities
How does HITL contribute to the CPS learning process? (e.g., how do humans verify own or others interpretation of sensor data?)
Do you innovate ways to enhance the humans in CPS with HITL? (e.g., use of remote experts, machine-assistance to the human in the CPS with HITL, other changes to human roles in HITL?)
How do you apply human-contributed knowledge in situations with ill-defined problems? (e.g., if a machine stops and there is metadata about its maintenance, how would you handle this knowledge?)
Do you have CPS with HITL with continuous interaction and data exchange between humans and machines, and where the machine technically has most of the control? (e.g., an autonomous vehicle with driver, other)

5.2.1. Socio-technical interactions and the human role

Socio-technical interactions and the human role were discussed in the interviews as the relation between product and customers and the moves to and from technical products and customers: ... *most of our installations are really close to the machines. You put in the drive [a variable speed control of electric motors] and then you don't touch it, right? You put it in and then you don't interact with it running. So there's not the same kind of consideration as you would have for a robot arm, for instance, where it moves around. ... Us, you install it, it's there. It needs to be accessible. But other than that, you don't usually care so much about it.* P1 Manager of research collaborations. From this quotation it is – not surprising – clear that industry experts see socio-technical interactions differently depending on the technology, i.e., the interactions are different for a drive (install, then no or only maintenance interactions) versus a robot arm (install and interact during use). However, this industry view is in stark contrast to ST theory that focuses on human users and use situations (Shao et al., 2019), trust in the interactions (Brauner et al., 2019), and wider interactions between different actors and intelligent components in physical world (Tuunanen et al., 2015).

The specific kind of socio-technical interactions called socio-technical co-agency in the literature appear in the interviews as discussions about equality between humans and technical things, how humans and machines feed into each other, the product plus the factory, engaging with social aspects when installing technical devices, and finally, as negative cases of co-agency that somehow touch on equality between human and machines. It is commonly stated across the different industry experts in the interviews that the aim is to remove the human from the loop: *...we don't think too much about how they interact. Humans, we think more about how will a machine interact with our simulation, such that a human can use them in the way they want... I actually think we're trending in some ways away from humans in the loop, in the sense that we're getting more and more data, we're getting more and more automation, we're getting more and more analytics done automatically. So I think it would be more and more the case that the humans are pushed to the very top of the decision pyramid...* P1 - Manager of research collaborations. *So, I'm seeing it in the context of, so, as for an example, when a robot is moving, it is generating a ton of data, and you get data from the robot, but you also get a lot of data from all of the other components in the work cell, and that's usually the system, right? So, the human touch of this, from our point, will usually be the, you could say, the operator of the robot. And, in the perfect world, there should be no intervention, right? But what we know is also that when you have an application running or a work cell running, it's about keeping it running with as little human interaction as possible.* P2 - Head of Product Development. *...of course, we are re-arming the machine or we are re-arming the processes in that way or another, then, of course, a human figure is necessary...usually there is somebody who is called a super user. Super user is somebody who probably will have a bigger experience if it comes to the machine, was trained in the best way, was the first, probably, operator on that machine, and will be somebody who is... who will be the first to contact if something is happening.* P3 - Project Manager. *The whole idea is automation, right? And that is removing the human from the loop...it's fairly simple, right? You know, the reason we put robots into manufacturing tasks is because then we, you know, avoid human errors.* P4 - PhD fellow and Specialist. The different industry experts P1–P4 express in these quotations that they view CPS as a separate entity from the human in the HITL. This is in strong contrast to the literature, e.g., [Fass and Gechter \(2015\)](#), which calls for the relationships between CPS and the humans being named and called out as important.

Another kind of socio-technical interactions, socio-technical CPS networks appears in the interviews as talk about both technical and human/social parts and their relation plus their varying positions in the networks: *...for instance, the water supply company...we need to support the operator going and looking at the drive. But we also need to support the operator running a big [...], controlling hundreds of drives in the factory where they don't interact directly with the drive. They really only care about, let us say, the belt going a certain speed. And the drive is just the middleman in that setup.* P1 - Manager of research collaborations. Here the interview data show how technical devices may be positioned as “middleman” in the setup, linking different humans to different other devices and to other groups of humans. Interestingly, this positioning of technical devices as social network nodes in multi-agent systems is the mirror image of the ST literature, which positions the humans in these roles ([Kampik et al., 2022](#)). Furthermore, the interview data indicate the variety of human specialists involved in the socio-technical CPS networks: *...we will much more consider software people in the process. And the people who will be responsible for simulations, which are different from the people who are developing protocols, who are different from the people in considering analytics, who are different from the people augmenting stuff with AI, right? So there will be this chain of people we need to think about more...* P1 - Manager of research collaborations and *...end users. And that's a very it's a very so it's a limited way of having a scope when you take a look at things because the ecosystem is obviously much bigger than that...* P4 - PhD fellow and Specialist. The industry experts see these

socio-technical CPS networks as including not even or not only end-users but a variety of actors – different types of software professionals – in “chains of people” and “ecosystems”. This presents a nuanced understanding of human ‘nodes’ in the network that goes beyond the presentations of ‘users’ or ‘humans’ in socio-technical CPS networks in the ST literature, e.g., [Cimini et al. \(2020\)](#).

In contrast to the literature ([Kampik et al., 2022](#); [Sadiq et al., 2022](#)), however, industry experts don't see a need for governance in the form of norms, policies, and preferences, such as responsible and agile approaches to information use. Norms and policies for behavior (e.g., safety-related) are not needed because the industry experts focus is on the technical devices, and, as stated by P4 - PhD fellow and Specialist., “robots don't wear shoes”.

5.2.2. Cognitive capabilities

Cognitive capabilities is discussed in the interviews as negative cases (no cognitive capabilities): *The short answer is we don't* P1 - Manager of research collaborations. Obviously, this response is far from the ST literature's suggestion to study techniques to address the cognitive capabilities at different levels in the socio-technical system ([Gil et al., 2020](#)).

The notion of amplification of machine learning (ML) by humans from the literature, is discussed in the interviews as when human (customers) amplify the machine, e.g., by feeding it data: *If there is, for example, long-term initiative to increase the productivity of all lines, then, of course, some initial measurements... Initial measurements are taken if what part of the production line you need to involve in, you need to put...* P3 - Project Manager. The industry expert in this quotation is aligned with the ST literature insight that HITL may extend ML capabilities by allowing HITL to contribute to the learning process ([Emmanouilidis et al., 2019](#)).

Human-contributed micro-knowledge, which in the literature is when data (in the form of micro-knowledge) are provided to the machine by humans, is discussed by the industry experts as user-feedback in software design, nudging, and the impossibilities for humans to contribute to the CPS processes: *We are launching a new programming, a new user interface this year. And one of the important things that we've done is that we have been doing it with customers. And having reference applications where you agree with someone to have this installed as the solution for your specific work cell, and then getting close, having a close encounter and getting close feedback on what works and what doesn't* P2 - Head of Product Development. In this quotation, the human-contributed micro-knowledge to the CPS is discussed as feedback to design when designing new features (e.g., new use interfaces). Furthermore, nudging plays a role: *...those actions are usually defined by the users. They want the systems to nudge what they are already trying to do and what they do every day to create value for their own business.* P2 - Head of Product Development. Here the human-contributed micro-knowledge to the CPS is for setting up systems. In contrast, industry experts also point out the challenges for human workers to contribute: *...very often it is impossible to catch those very small changes that you need to do on the production line or in the structure that you are working that can be automatized. This is a very, very often that we are talking about very complex processes, very complex situations.* P3 - Project Manager. Here the potential human-contributed micro-knowledge to the CPS is to the real-time running of the process. Which according to the interviewed expert is not really possible. Furthermore, the industry experts see the human contribution as a contribution by educated experts rather than by the workers (end-users): *We organized sprints with people just to have like...to throw a lot of people on one line or on one process just to figure it out how it is working, how to do it better, how to work with that... It could be once per six months. Somebody or some group of people that are highly educated or know what they are talking about and they are just finding new potential solutions, ideas, something like that. Without us...* P3 - Project Manager. Here the human-contributed micro-knowledge

to the CPS is innovative process or line micro-improvements. Overall, regarding human-contributed micro-knowledge the interviewed experts suggests that human-contributed micro-knowledge to the CPS can be design feedback, configuring system settings, real-time process changes (usually not possible to catch by humans), and process or line micro-improvements. None of these clearly related to the situations with ill-defined problems discussed in the ST literature such as in Emmanouilidis et al. (2019), but appear to be well-known routine problem situations in companies.

Another issue from the literature is how to enhance human capabilities by machine generated protective stops, also those over time indicating need or maintenance. The response from the industry experts differs: *...at some point the operator is going to say it doesn't work, let me send it to repair. And that's how that works, right? But with the safety, we do all of this automatically, and that's kind of where we are. We're trying to automate these things.* P4 - PhD fellow and Specialist. This quotation shows that in contrast to the literature (Emmanouilidis et al., 2019), which see CPS-enhancements of human capabilities as giving humans a bigger role to play, the interviewed industry experts see the human role diminish at least for safety related issues.

A joint cognitive system is another notion from the literature that was discussed in the interviews: *A joint cognitive system is one that enhances the cognitive capabilities of the whole socio-technical CPS with HITL where we have a collaborative space between the operator and the robot. And those places are, you may say, developed not really to work with the process itself, but work to build a machine that is only supportive for the process. And the operator is only supportive for this setup.* P3 - Project Manager. It is clear from the strong focus on the machine rather than the process in this quotation that the interviewed industry expert is a provider of CPS and not the end-user company.

5.3. Interviewees' response to human-centered design

This section presents the industry experts responses to interview questions on human-centeredness, Table 4. Note, it does not directly address human-centered design, as this was better discussed across themes.

5.3.1. General psychology of CPS with HITL

The notion of a sensing and actuation role of humans from the literature was not obvious to the interviewed industry experts. They did not see possibilities for co-sensing of machines and humans, but rather delegated the human to breakdowns and emergency situations: *...we've implemented acceleration sensors, force torque sensors and temperature sensors and joint and end effectors, and we can obviously use those. Yeah. The human is more for you know...something structural ... if you fuck up the roof... I don't know.* P4 - PhD fellow and Specialist. This positioning of the human in a role outside the sensing and actuating activities is in contrast to the literature (Cimini et al., 2020) that see the sensing and actuation role of human users in CPS with HITL a maybe enhanced by CPS due to their physicality (Cimini et al. 2020). In addition, human needs for sensing and actuating appeared in the interviews as discussions about actuating, that is, when there is a need for human agency and consumer agency. This included that humans have need for doing something e.g., making decisions, what are human need-definitions for sensing and actuating, and also what are good business strategies based on human needs for sensing and actuating. Finally also age or expertise were discussed as influencing factors on role of humans as sensors and actuators.

A related issue, human multi-sensory perception and action is according to the ST literature shaped by, and can be supported by, the CPS with HITL. The industry experts saw it from a slightly different point of view: *say, okay, which tasks actually need really high latency. Then we do that without the Human-in-the-Loop. The robot is made such that it cannot do anything that would break the safety requirements and safety requirements have, you know, really, really low latency. Uh, close*

Table 4

Interview questions on Human-centeredness, derived from Section 4.2.

General psychological phenomena of CPS with HITL
How does CPS with HITL enhance human users sensing and actuating possibilities (e.g., what can they do which they could not do before?)
What differences of humans in CPS with HITL do you consider? (e.g. expertise, age-related needs, other?)
What factors for human multisensory perception and action do you consider? (latencies video/audio/haptic, biases, health, economic conditions, other?)
Group-centered CPS with HITL
which human control models do you apply? (e.g., human reliability models?)
How do you deal with tight latency requirements for multisensory scenarios? (e.g. allow humans to act bodily in digital space as if they were in physical space)
How are humans and other subsystems/intelligent agents able to continuously integrate into the coherent CPS with HITL? (e.g., what capabilities to continuously assess and maintain their own integration status within the overall system composition do subsystems have?)
Individual-centered CPS with HITL
Are you aware of human biases and how they may hamper CPS with HITL? (e.g., overconfidence, misinterpretation of feedback event sequences, other?)
How do you place the human at the center for all design and verification processes of CPS with HITL? (e.g., use of agile methods with feedback loops between consecutive stages for immediate and rapid testing and re-design)
How do you do human reliability assessment of CPS with HITL? (e.g., use simulators?)
Do you support experts in transferring skills from human to machines, or reversely, enable expert task performance by novices following instructions and guidance from the machines?

to none, actually. P4 - PhD fellow and Specialist. The industry expert in this quotation is focused on the task and the technical device and how latency is required for these, while the ST literature is focused on how human perception and action are shaped by human-sense-specific latencies in CPS with HITL that calls of other non-technical performance metrics (Fitzek et al., 2021; Gopalakrishna et al., 2017)

5.3.2. Group-centered HITL

Group-centered HITL for the industry experts were not so much about the ST literature's ideas about human control models for groups or human-machine networks, but rather about the human teams. Human team models were discussed in the interviews as matters of organization, SCRUM principles, different human teams in the organization, sharing, and a focus on the people rather than the work: *We do a lot of talking to our customers. And we do have a lot of, some of our big core customers, we have good interaction with. We talk a lot with them. And it is a lot driven by customer features, right? And our whole development organization is built around agile principles and Scrum's principles with a lot of interaction with user stories and pulling in what we're going to do.* P1 - Manager of research collaborations. Here in this quotation the human team model in CPS with HITL is an agile software development team. In the literature, the idea is that humans need data and visualizations to become aware of interactions between artificial agents and the way these are communicated to the other system constituents (Botev, 2020); the industry experts' pointing to agile teams is perhaps somewhat related to the literature insights here. Besides of that, the industry experts express some hesitance towards the idea of teams: *...there are examples of multiple operators working on or with one robot. But I think from our part, it's important that you are able to do it alone. You can only operate the robot with one hand when you're free driving, right? So, I can't think of anything specifically where we have thought into the development process that this needs to be able to do with more than one operator.* P2 - Head of Product Development. In this quotation, the human team model in CPS with HITL is seen as an add-on for specific situations, and the industry experts have to speculate about such situations. The literature insights on this issue are not so different from the industry

viewpoint, in that the literature is nuanced about how to co-adapt models of humans for CPS with HITL in terms of heterogeneous human populations with different issues of access (Fitzek et al., 2021). The industry experts see the nuances: *We will need to set the machine in a certain way with certain tools and in certain parameters. And so, there is a book of knowledge how to do it. And of course, there is this part of a human operator that will know better a little bit instead of reaching to the book.* P3 - Project Manager. Here the industry experts are pointing out the human workers' general abilities to contribute, which the ST literature discusses as the idea that subsystems (e.g., human and software/hardware agents) are equipped with capabilities to continuously assess and maintain their own integration status within the overall system composition (Botev, 2020).

5.3.3. Individual-centered HITL

Individual-centered HITL issues mentioned in the literature, such as issues of rational human behavior and digital human tactile skills towards the physicality of the CPS, were not discussed by the industry experts. They had more to say about verification and reliability.

Having the human at the center of verification is presented in the literature as a good idea. The human must be a persistent center for all design and verification processes of CPS with HITL. When asked about this, the industry experts talk broadly about safety, human understanding, and specific work domain cases: *...the safety is set up is, of course, with humans in the loop, in the sense that there is a choice of how these safety systems should work, but they cannot wait for human interaction, right?* P1 - Manager of research collaborations, and *...at the end, you should at least understand what is happening.* P3 - Project Manager, and *...in remote surgery, for example, you would have the human before the software, and then the software would take the human inputs as input and treat them and execute something with the hardware surgery. And so, so, so in any case, where in any case where whatever the robot is doing is directly related to the human. Yeah, I guess that would need to be some sort of interface for it.* P4 - PhD fellow and Specialist. In these quotations, the industry experts somewhat reluctantly agree with the overall idea of having the human at the center of verification of CPS with HITL. However, the detailed issues present in the literature, on human-centeredness in verification of all subsystems (Cuckov et al., 2017) and increased safety by expressing sensor information “linguistically” rather than as quantities (Banerjee et al., 2020), do not appear in the interviews.

Reliable human behaviors in HITL was clearly an issue of interest for the industry experts: *...people are trained for certain different situations, a little bit like in the army, you may say. When the situation is, of course, appearing after some sort of delay, then everybody after a second, third delay will know what more or less was the delay. So they are more or less know what to find out after a while and how to prepare themselves for that. So this is, and due to the, of course, all the time changing environment people are training themselves even quite fast.* P3 - Project Manager. The industry expert in this quotation appears perhaps more optimistic about human reliability of CPS with HITL since they state that people are studying their task performance within various operating scenarios and evaluate it to learn from it. In contrast, the ST literature points out that information about behavior related to the observed CPS with HITL task is often unavailable (Jirgl et al., 2019).

5.4. Interviewees' response to human-computer interaction tools and methods

This section presents the industry experts responses to interview questions on human-computer interaction, Table 5.

Table 5

Interview questions on Ergonomics, derived from Section 4.3.

Human factors
How do you refer to the human in CPS with HITL? (human factor, human actor, human decision maker, human operator, other?)
Human-computer interface
Which human-computer interfaces for CPS with HITL do you have? (e.g., <to come, build up a list>)
Human-machine interface
How do you ensure that the human-machine interfaces support changes in HITL-users' situational context and minimize human attention costs? (e.g., how do you capture and analyze what happens in the human-machine interface to detect human errors and support human performance in real time?)
Ergonomics of CPS with HITL
What are the ergonomic issues related machine and human sensing for safety in CPS with HITL? (e.g., robot cages, proximity-sensors, intention-inferring, use of non-human senses such as Lidar, use of surveys and prompting of employees, other?)
What are the ergonomic issues related to adjustment of CPS with HITL over time as the human skills evolve? (e.g., micro-learning opportunities, easy programming, ML-generated info-displays, other?)
What are the ergonomic issues related to the provision of some human autonomy in CPS with HITL? (e.g., providing portable manually operated measurement devices that support the human engaged in quality and process control decisions, other?)

5.4.1. Human factors

Human factors views humans as a system component, which in the CPS with HITL literature is presented as equally important to the system as other components such as software and hardware, and which may include engineering, monitoring, controlling, and performing operational maintenance roles. The industry experts tend to agree with this view, but also wants to eliminate the human factor from the loop: *In the traditional usage of our drives, mostly they are systems that are set up to control a process. Let us say whatever, a conveyor belt or something. In the setup, there is a human interaction in the sense that they will need to commission some of it and set it up. And then for the rest of the time, they will usually be part of some scatter control system. And then there will be interaction that way around, right? So that's the very traditional approach. As we go towards more intelligent products, this is beginning to shift a little bit in the sense that we are looking at situations where, for instance, something like predictive maintenance, in the sense that the system will automatically suggest maintenance schedule or monitoring and these kind of things. And then humans will be informed, right?* P1 - Manager of research collaborations. The industry expert argues that the human will be informed directly about faults that are sensed by the CPS, and seen in this way, this is also what the ST literature talks about when discussion how the human becomes central in the design of CPS with HITL because of the movement from the notion of information to one of sensing and actuating and with human and machine operators seamlessly connected (Fitzek et al., 2021). On the other hand, the industry expert is arguing that the massive amount of data shift the human to higher positions in the control system and states: *...in some ways, we are moving humans away from the loop, but then we are putting them at the top of this cognitive pyramid* P1 - Manager of research collaborations.

5.4.2. Human-computer interfaces

Human-computer interfaces or Human-machine interfaces issues mentioned in the literature were in the interviews discussed as interfacing or interfaces, with human-computer interaction being not so much about machines, but rather about computers with very developed/complex software and resulting user interface (UI) considerations: *...the individual drives don't have any interface. You interact with them only via a computer or a setup or PLC system. And of course, we need to consider how do you then interact with it and how can you support these*

various modes of interactions. So, there is a lot of UI consideration and interactions considerations. P1 - Manager of research collaborations, and *We are launching a new programming, a new user interface this year. And one of the important things that we've done is that we have been doing it with customers.* P2 - Head of Product Development. The industry experts are aware of the need for user interfaces somewhere in the systems and also that these should be developed with customers and with many considerations. This is not surprising, since these viewpoints emerged in HCI research and practice in the 1980ties, and for human-machine interfaces even long before (Singh and Mahmoud, 2019). In these quotations, the industry experts are also aligned with the ST literature that says that today human-machine interfaces should support changes in user situational context and minimize human attention costs (Shao et al., 2019).

5.4.3. Ergonomics of CPS with HITL

Ergonomics of CPS with HITL issues from the literature came up in the interviews mostly related to safety considerations and adjustment over time of CPS with HITL. Human autonomy which is important in the ergonomics literature on CPS with HITL, was not discussed by the industry experts.

Ergonomic methods appeared in the interviews as (much) talk about safety for humans or privacy related to ergonomics, safety cages, labor safety standards, and how to use the product/machine/computer: *...as a very concrete example, when we have, for instance, textile mills, where they have these very big rollers running, and you have a power outage, which happens a lot in India or other countries where the grid is not very good, you have to do something very quickly, right? Or you will destroy the roller, you will simply destroy the factory if you don't do anything, right? So these kind of safety features cannot wait for humans to make a choice, right? The choice will be made, right? And maybe you can decide up front, in some cases, what choice would you like to be made, but the choice will be made, right? And in many cases, the choice will be, we will destroy the drive to save the system, right? We will simply take too much into our capacitors, and we will destroy the drive, because that's usually the cheapest component of the system, right? ... And I think it is a valuable understanding for humans in the loop that systems that are fast acting and dangerous, you cannot be in the loop, right? You can be in the provisioning, but you cannot be in the loop...* P1 - Manager of research collaborations. While the industry expert in this quotation clearly talks about safety, some considerations about economy are also there. Other industry experts also expressed economic concerns together with safety and ergonomics: *...by having a robot where it's actually safe to stand beside, then you actually also open up for automating a lot of new things* P2 - Head of Product Development, and *...if you have a space, you are able to build another factory. If this will give you benefits. So all of the topics, all of the issues that you will find out from the ergonomic point of view are very much going to the lack of space.* P3 - Project Manager. Interestingly, while the ST literature on ergonomic methods talks about safety (Dani et al., 2020) and not economy, the industry experts bring in economy into the discussion of the benefits of good ergonomics, albeit indirect, in the form of cheaper cost, more automation, and space for more factories.

Another issues from the literature, how CPS with HITL that are adjusted over time, is part of the industry experts' views on the ergonomics of CPS with HITL: *...the issue is that if you will take a human operator from the loop, you will cancel or you will eliminate the human part of developing your idea or developing the process. You will only have a process that will work and you will always only have the machine, you may say. But adding to that part, the human figure, the human element, when people are trying always to tune in a little bit, work a little bit different, people are always trying to do something different each day or each month. And from that, you can, of course, develop some other solutions.* P3 - Project Manager. Here the industry experts are aligned with the ST literature, which says that it is important to adjust over time CPS with HITL to the humans skills by capturing changes in motor and cognitive abilities of the humans (Gallina et al., 2015).

Table 6

Interview questions on human work , derived from Section 4.4.

Interaction designs for work
Does interaction designs for work follow push and pull principles so that both sides can request resources, and does this depend on the work task's criticality? (e.g., can the computer signal the need for additional labels as a data modification from the human, can the human request information and operations from the computer, other?)
Work design
is the design of the manufacturing process and the human work design done together? (e.g., is the design of the data labeling, labeling workers' knowledge, organization of the labeling work, and suitable technical tools, done together?)
How do you innovate work design for novel types of work systems? (e.g., consider extreme scenarios in terms of type of work task, isolation from other human societies, and extreme physical settings; task adaptability, asynchronous communication, shortage of specialists for CPS with HITL work design, specific stress conditions, other?)
Work environment
how do you ensure that humans are in the loop and in control of the smart work environments? (e.g., CPS with HITL for climate management, light, security registration, energy conservations, other?)

The industry experts however do not consider potentials of smart human interfaces with micro-learning opportunities, easy programming, and AI-generated information to over time enable task, context, and user-adaptive interactions (Brauner et al., 2022).

5.5. Interviewees' response to CPS with HITL in human work settings

This section presents the industry experts responses to interview questions on human work design for CPS with HITL, Table 6.

5.5.1. Interaction designs for work

Interaction designs for work notions from the literature are responded to in the interviews by giving examples of interaction designs and how humans interact with the system. The industry experts pointed out that there are more and more HITLs due to changes in interactive technologies such as virtual reality (VR), digital twins, and haptic interfaces: *We have more and more humans in the loop in the sense that we provide more information to the operator, right? ... [then] there's the maintenance aspects ... we are beginning to look at doing, for instance, these kind of VR solutions. I don't know exactly how they fit into Human-in-the-Loop? ... then we have ... digital twins and these kind of things. And I'm not exactly sure how to think about Human-in-the-Loop in these setups...* P1 - Manager of research collaborations, and *...we have looked into stuff like haptic feedback, like a vibration. Because when you program the robot... you're holding the robot. And you can have some kind of haptic feedback through a vibration in the robot. And it can give you feedback on a certain point that you're in, or a certain position.* P2 - Head of Product Development. In these quotations, the industry experts view are aligned with the ST literature which argues that interaction designs for CPS with HITL are realized towards new forms of smart work environments in which the boundaries of virtual and physical worlds dissolve and novel forms of interaction are applied (Magerkurth, 2012). The industry experts in these quotations do not speak about moderating factors on the interaction design (work task's criticality, the experience of the users, organizational issues (Zschech et al., 2021).

5.5.2. Work design

Work design for CPS with HITL is a notion from the literature that is responded to in the interviews as being about work design for a company, such as manufacturing processes and the overall work processes. Furthermore factory floors and production lines may have related social issues and innovations in work designs, with utilization of industry standards and norms and with issues related to company strategy and sustainability goals. Work design is also discussed in

relation to building a factory, which is something industry experts frequently are part of or expects to be parts of in the very near future. This may include HITL and operator role in virtual reality and similar cyber-systems, which increase the awareness that the operators work will have to change.

Regarding work design for CPS with HITL, however, industry experts in various ways doubts the value of HITL: *...if you build a factory and you need industrial automation, you know drive systems. And it's more a question of you looking at the manuals, okay, how do I get this feature with the [company name] product as opposed to the [name of competitor] product? But you will expect the feature to be there...there is a situation here where there is more domain knowledge from <company> and in the drives than the human have, and here we will sometimes remove the human from the loop, right?* P1 - Manager of research collaborations, and *...if you are a machine technician, you're not necessarily skilled enough to reprogram a robot...that is a part of our platform, that is part of the robots, not necessarily part of the work cell.* P2 - Head of Product Development. Here the industry expert P1 mentions situations of providing their system to clients and P2 talks about their product as separate from the 'work cell'. These industry expert views contrast the ST literature that argues that to achieve high quality work design, the design of the manufacturing process and human work design should be done in a socio-technical way, that is, designed together (Schuir et al., 2021). On the other hand, the industry experts also see possibilities for co-design of CPS with HITL as suggested by the ST literature: *... if we make it even easier to reuse the old program and make the reprogramming intuitive enough, then it actually improves the business case for a lot of new production setups that again generate a higher need for automation.* P2 - Head of Product Development, and *...if you look at us historically, that was one of the biggest breakthrough when introducing our robots, is that you could actually move the robot around and program it through waypoints instead of having to type in abstract code, you could say.* P2 - Head of Product Development, and *...part of the production line is automatized, it is a good procedure to bring them out, the operators, into the machine, into the process of building a new machine. So, that their knowledge will...will be copy-paste.* P3 - Project Manager. The industry experts are clearly aware of the issue of co-design.

However, about *approaching the customer and clients* and end-users, the industry experts explains that from a technology provider perspective, the co-work-design is of lesser importance: *No, it's not our main thing, right? Of course, we do it to some degree in the sense that you can get support, you can get material, you can get manuals, you can get some hands-on training. But it's not like, it's not a big part of it.* P1 - Manager of research collaborations. These negative approaches to co-design contrasts the literature which is positive towards co-design of work for CPS with HITL.

Innovative work design, which was an issue in the literature, is discussed in the interviews as how things work with the product, and what are the work processes with all their relations to workers, customers, etc. Furthermore, it is discussed as innovation in the expert's own company and from customers, innovation difficulties, and innovation and higher levels of automation. Finally, innovative work design is discussed in the interviews as work design with AI or with agile work principles. The industry experts pointed to limitations for radical innovation, including client conservatism, that innovation has to happen at the clients, and that they themselves is unsure about how to innovate work designs for CPS with HITL: *...the industries we target are usually pretty conservative. Industry is pretty conservative, logistics is pretty conservative in the sense they don't want big changes.* P1 - Manager of research collaborations, and *...when we move to more data intensive industries and Industry 4.0 and these kind of things, right? It will be much more the case that we provide features to be part of the orchestration of a smart setup, but we will not do it ourselves.* P1 - Manager of research collaborations, and *...we provide the platform, being the robot arm, and then we sell through distributors and usually we always, also always have an integrator involved. And the integrator is the one setting up the solution*

with the customer and buying not only the robot, but also every other component and having the system work as a whole, so in the work cell. P2 - Head of Product Development, and *...when you introduce something like that, a new technology on your workplace, it's almost always the case that it is creating some uncertainty. Because it is addressing how we are doing things today versus how we should do it tomorrow. So, that is a theme, yes. And it is something that we also, I think, I believe, we need to learn even more about how to address that.* P2 - Head of Product Development. These quotations illustrate how industry experts' in this case rather conservative viewpoints are quite far from the ideas in the ST literature, which are about a variety of novel work systems (Brauner et al., 2019; Cimini et al., 2020) that can be designed by considering extreme scenarios (Linder et al., 2022).

5.5.3. Work environment

Smart work environments mentioned in the literature appeared in the interviews as discussions about 'work' plus something else such as autonomous systems, digital twins, IoT, AI, or selfcontrolling systems. In addition, there were discussions about smart work environments in relation to smart factories, which concerned reducing the role of the human, and removing HITL and lower level cognitive functions. Finally, there were discussion about the general plan for advanced technology as it is and the 'automation vibe': *...these kinds of systems will become much more just a commodity. You will pick whatever off the shelf as long as it supports the kind of data you want to interact with. And to be successful, this is the game you need to play, right? You need to play nicely with other data systems. You need to play nicely with automation. You need to play nicely with smart features. And this will be the differentiating feature. And that is something that goes much deeper than just the machine-human interface* P1 - Manager of research collaborations. In the quotation, the industry expert talks about the environment of the machine, not the work environment of humans that the ST literature is about. The literature suggest that CPS with HITL could include automation and centralized control systems that provide comfort (Gopalakrishna et al., 2017), security registration, other services, and conserve energy, and, furthermore, by giving humans control over data relevant to their work environment is known to positively influencing the productivity (Loengbudnark et al., 2022).

6. Discussion

The objective of this paper concerned expanding the CPS with HITL terminology, which we did by systematic reviewing research on CPS with HITL from socio-technical perspectives. The insights gained were validated for their usefulness to industry and potential areas for industry-oriented research by interviews with industry experts and by systematic discussion of the socio-technical perspectives. The paper contributes to fill the gap that exists because of a lack of CPS-specific comprehensive and practical guidelines that consider HITL from socio-technical perspectives in environment, infrastructure, coordination, and decision making with CPS.

6.1. The socio-technical perspective(s) on CPS with HITL

Our four research questions were about socio-technical capacities, a human-centered approach, human-computer interaction technologies, and human work design. Table 7 maps findings for each question to the four main areas of CPS with HITL adaptations listed in Section 2: environment, infrastructure, coordination, and decision making.

Table 7
Socio-technical perspectives on research into CPS with HITL adaption.

Research into CPS with HITL adaption				
<i>Socio-technical perspectives</i>	Environment	Infrastructure	Coordination	Decision making
<i>Socio-technical interactions</i>	Human-algorithm interactions	Socio-technical CPS networks	Socio-technical co-agency	Cognitive capabilities
<i>Human-centered design</i>	Evaluating reliable human behaviors; Sensing and actuation role of human	Human at the center of verification	Human team models	Human needs for sensing and actuating; Human multisensory perception and action
<i>Human-Computer Interaction</i>	Ergonomics of CPS with HITL	Human-machine interfaces	Human-computer interfaces	Human factors
<i>Human work design for CPS with HITL</i>	Human work environment	Work design for CPS with HITL	Innovate work design	Interaction designs for work

Note: Each cell content refers to subsections and/or paragraphs in the analysis chapter of this paper.

6.1.1. Environment and ST perspectives

CPS with HITL theory suggests that environmental contextual aspects such as physical, temporal, social, computational, historical, profile, and also humans are a part of the environment that interact with the CPS, that the associated constraints and limitations impact the adaptation requirements (Da et al., 2014). An ST perspective suggests to study Human-algorithm interactions between algorithms and human operators in key decision-making interfaces, and between algorithms and (company) social media users in small “filter bubbles” of reactions and likes (Valdez and Ziefle, 2018). The principle of human reliability from the ST literature suggests to study naturalistic human task performance and errors in CPS with HITL in operating scenarios (Jirgl et al., 2019). Thirdly, in dynamic CPS theory, handling context-aware adaptation by capturing and analyzing the context within the evolutive environment is complex. The ST literature on ergonomics of CPS with HITL suggests that it is possible to infer human intentions from past sensing information, by fusing information from multiple senses including some that humans do not have such as lidar and physiological signals (Dani et al., 2020). Finally, CPS with HITL theory argues that such systems are situated in the environment which most often include both physical and virtual elements (Weyns, 2017), which resonates well with the ST literature that suggests that workplaces are increasingly found within smart work environments provided by automation and connected services. However, the socio-technical perspective is that limiting or removing employees’ control over their work environment by automation with CPS result in dissatisfaction, and hence decrease productivity (Loengbudnark et al., 2022), while having CPS with HITL and hence human control is thus directly and positively influencing the productivity, given innovations in safe, efficient, and trustworthy CPS with HITL (Dani et al., 2020).

6.1.2. Infrastructure and ST perspectives

CPS with HITL theory states that the CPS *architecture* component determines variations in both software and hardware architectures, which are designed by stakeholders. The ST literature suggests to study such architectures as socio-technical CPS networks. These constitutes interactions between a variety of agents — humans, machines, productions, software, AI tools, and as such there among other things networks of social interactions (Cimini et al., 2020). Second, the theory states that infrastructures consist of various hardware (HW) and software (SW) component. An adaptive CPS *hardware* architecture can be re-structured in run-time to add, delete, replace, and combine its elements. The CPS HW elements include *sensors*, *network facilities*, *computation resources*, and *actuators*. The ST idea in having the human at the center of verification is that the human must be a persistent center for all design and verification processes of CPS with HITL, including design and verification of all physical, electro-mechanical, embedded systems, and the integrated high-level software executed on mobile devices, computing platforms, and cloudbased resources (Cuckov et al., 2017).

Thirdly, the CPS *architecture* component determines variations in both software and hardware architectures. These architectures are designed by stakeholders and self-adapted by the autonomic control element during system execution. The ST literature suggests that today there is a strong need for human-machine interfaces that support changes in user situational context and minimize human attention costs (Shao et al., 2019), which requires capture and analysis of what happens in the human-machine interface to detect human errors and support human performance in CPS with HITL in real time. Finally, a CPS might be in interaction with the dynamic surrounding environment by taking advantage of sense and actuate elements that sense elements frequently retrieve raw data Since the software system does not fully control the environment, the data transmission is continuous. Related to this issue of tackling the contextual dynamics, the ST literature suggests that the context of work design for CPS with HITL includes the type of production system in which work takes place, and hereunder task allocation between machines and humans, and the machines’ autonomy and the general level of automation in the work environment (Cimini et al., 2020).

6.1.3. Coordination and ST perspectives

In CPS theory, dynamic computation infrastructure, in coordination with the dynamic environment management, is the more general case where CPS belongs. In this space, both dimensions have to be managed, but keeping a separation of concerns with adequate coordination can help to face the complexity of the design. In the ST literature, socio-technical co-agency is when humans and machines/IT systems work together on an equal footing. CPS is seen as a separate entity from the human in the HITL. The relationship between them is named or at least called out as important (Fass and Gechter, 2015). Humans are autonomous and cooperating conceptual entities that embed the interface needed to communicate with machines (Cimini et al., 2020). The resulting new system that includes both humans and machines is a “socio-technical CPS” (Cimini et al., 2020). Second, in CPS theory, co-ordination in embedded systems with specific computing capacity and low energy consumption happens to adapt to e.g. changes in the control system itself (e.g., fault recovery). In the TS literature, the idea in human team models for CPS with HITL is that the actions and reactions of the humans play a central role to self-awareness, self-adaptation and self-integration of the various artificial agents in the system, and the human on their side need data and visualizations to become aware of interactions between system constituents (Botev, 2020). In addition, to co-adapt models of humans for CPS with HITL, the HITL component is also there to ensure the well-being of the human operators (Fitze et al., 2021). Third, in CPS theory, the coordination is carefully analyzed and addressed mainly through distributed embedded CPS devices that are distributed across the system. In the ST literature, the human-computer interface is a specific material component in the computer system in the same way as the digital, analogue, and cyber components (Cuckov

et al., 2017), which then becomes part of the coordination. Finally, hard real-time systems requiring strict predictability is used in very dynamic environments like physical processes control (e.g., engines, signal processing). In the ST literature, innovative work designs are suggested for extreme process control scenarios (Linder et al., 2022). The human work design technology itself may need to be further developed to design for a variety of novel work systems (Brauner et al., 2019), such as novel hard real-time systems.

6.1.4. Decision making and ST perspectives

In CPS theory, users prefer proactive adaptation because of its ability to avoid quality degradation within the system. However, the proactive feature requires running complex prediction algorithms that depend on the correctness of entry data as parameters. The ST literature suggests that the amplification of machine learning by humans, for example in condition monitoring, may add human expert knowledge to data driven capabilities (Emmanouilidis et al., 2019; Banerjee et al., 2020). Second, adaptation policies depend on the time of the system's reaction to uncertainties, i.e., before or after risk occurrence. Here for example the time aspect and IoT system characteristics and goals impact the strategy and decision method established for adaptation, and the ST literature suggests to add that human multisensory perception and action needs require adding new performance metrics for CPS with HITL, such as psycho-physical and psychological (Gopalakrishna et al., 2017). Third, ST theory argues that the human decision maker is a risk factor in CPS with HITL, so it is important to design support for human decision makers' trust and compliance (Brauner et al., 2019). Finally, much research focuses on reactive adaptation. The monitoring and execution activities are very much the same in reactive and proactive methods, but the analysis and planning phases make the difference. The ST literature suggests that designing interactive prototypes for work can help to improve decision making in CPS with HITL (Gil et al., 2020), e.g., by providing novel forms of interaction (Magerkurth, 2012), and by pointing to moderating factors on the interaction design such as the work task's criticality and the competence of workers (Zschech et al., 2021).

6.1.5. Lessons learned

The discussion above of the findings from the literature review teach us that areas of concern for CPS with HITL adaption can successfully be viewed from four different ST perspectives. There is therefore a possibility for better theorizing CPS with HITL by integrating these perspectives in new control models. Perhaps these models need to be work domain specific, e.g., different for driving and for manufacturing etc. Most importantly, the idea of CPS with HITL is that the human is continuously, in real time, a part of the CPS and contributes to the outcome.

An important lesson from this study is however that the potential of ST perspectives on CPS with HITL is strong in the literature, but it is weak or ignored by industry experts, Table 8. The industry experts who we asked to respond to the insights from the literature review, tend to agree most clearly with the ST literature insights regarding the importance of classic HCI. There is also some agreement about human team models and reliable human behavior and their contributions to verification of CPS with HITL, Table 8. Agreement with socio-technical perspectives on socio-technical interactions and human capabilities, general psychology, and work and interaction design are less, Table 8. However, since there is some overlap between the literature insights and the industry experts views, these, these topics are a good place to start for industry-oriented research into CPS with HITL and from ST perspectives. There is a need for positioning of humans and technical devices in the CPS with HITL networks, bring in the rich variety of ideas in ST literature on how to enhance HITL, embrace a ST terminology, and do detailed field studies of manufacturing workers and their interactions with a full range of humans and technical devices in the CPS networks.

Table 8

Alignment of industry experts' view with literature insights.

Interviews	ST Literature
	Socio-technical
D	Socio-technical interactions
?	-Human-algorithm interactions
D	-Socio-technical co-agency
DA	-Socio-technical CPS networks
D	Cognitive capabilities
A	-Amplification of ML by humans
D	-Enhance human capabilities
D	-Human-contributed micro-knowledge
D	-Joint cognitive system
	Human-centered
D	Psychology
D	-Sensing and actuation role of human
?	-Human needs for sensing and actuating
D	-Human multisensory perception and action
A	Group-centered HITL
?	-Human control models for groups
?	-Human machine networks
A	-Human team models
A	Individual-centered HITL
?	-Rational human behavior
A	-Human at the center of verification
A	-Evaluating reliable human behaviors
?	-Digital human tactile skills
	Human-Computer Interaction
A	Human factors
A	Human-computer interfaces
A	Human-machine interfaces
A	Ergonomics of CPS with HITL
AD	-Ergonomic methods
A	-Adjusted over time
?	-Level of human autonomy
	Human work and CPS with HITL
A	Interaction designs for work
AD	Work design for CPS with HITL
D	-Approach to the customer
D	-Innovative work design
?	-Industry norms and expectations
D	Human work environment
D	-Smart work environments

Note: D = Disagreement between interview data and ST literature; A = Agreement between interview data and ST literature; ? = No interview data.

Finally, there is a need for questioning scenarios for when to use CPS with HITL and when to use CPS with full automation. The literature sees a strong potential for development of the HITL part, while the industry experts in our study are perhaps more conservative and want to remove the HITL wherever possible and go for full automation. Thus there is a need for 'ambassadors' for the HITL part if it is going to be used.

In summary, based on the lessons learned, we provide recommendations for software engineering researchers, Table 9. These recommendations aim to bridge the gap between academic insights and industry practice, fostering research that is both theoretically rich and industrially relevant.

6.2. Implications for professional practice

Based on insights discussed above and on the general ST literature, we propose the practical solution that producers, vendors, and customers/end users significantly enhance their co-design activities. As our expert interviews indicated, there is increasing awareness among producers of the importance of going beyond the simple scenario that the vendor installs the equipment and then the customer operates it. One of our interviewees stated that "... one of the important things that we've done [this year] is that we have been doing it with customers...having

Table 9
Recommendations for software engineering researchers.

Recommendation	Explanation of recommendation
Explore the Overlap	Focus on areas where there is an overlap between literature insights and industry expert views, particularly in classic HCI 4.3, human team models 4.2.3, and reliable human behavior 4.2.4. This common ground may yield practical and impactful research outcomes.
Embrace Socio-Technical (ST) Perspectives	Incorporate ST perspectives into research to enhance Human-in-the-Loop (HITL) in Cyber-physical systems (CPS). This includes adopting ST terminology and conducting detailed field studies on the interaction between humans and technical devices within CPS networks 4.1.
Develop Domain-Specific Models	Recognize that CPS with HITL can be viewed from multiple ST perspectives. Use this diversity to create new control models that are tailored to specific work domains, such as driving or manufacturing 4.2.3.
Continuous Human Integration	Emphasize the importance of real-time, continuous human participation in CPS 4.1.2. Research should investigate how human contributions can be optimized within these systems.
Balance HITL and Automation	Challenge the industry's preferences towards full automation. Advocate for the value of HITL where appropriate and research scenarios where HITL can complement or outperform full automation 4.4.
Act as HITL Ambassadors	Encourage researchers to become 'ambassadors' for HITL, promoting its benefits and potential for development, especially in areas where industry may be hesitant to implement it 4.2.1.

Note: The numbers in parentheses are references to sections in the literature review.

reference applications where you agree with someone to have this installed as the solution for your specific work cell, and then getting close, having a close encounter and getting close feedback on what works and what doesn't..." P2 - Head of Product Development.. Thus the industry expert is aware of the need for co-design with the workers in their work cells. The insights from the literature review supports increased use of human-centered design that includes more and more nuanced views of the humans in the HITL, see e.g., Section 4.2.1. Professional practice can draw on the existing richness of frameworks for the practical ST design solutions; for example variants of professional practice-based design or human work interaction design (Clemmensen, 2021).

6.3. Implications for future research

The insights from our expert interviews, which showed industry experts' awareness of issues with CPS in HITL in industry due to more complex automation, see e.g., Section 4.1, indicate that there is a potential need for identifying concrete examples of successful socio-technical integration in industry settings. Noteworthy, the literature review indicated that giving humans control over data relevant to their work environment is positively influencing the productivity (Loengbudnark et al., 2022). From an ST perspective, such studies could start with studying human workers' positive practices and positive experiences (Klapperich et al., 2018). Potentially there is a need for analysis of CPS with HITL successes within the range of work domains and application domains.

Our study of the software engineering literature identified four main socio-technical approaches to CPS with HITL-related social science. There were few if any traces of the possible emerging shift from existing human-centered approaches towards current trend of human-AI collaboration, but see e.g., Section 4.3 and Fitzek et al. (2021). Generative AI may however change CPS with HITL radically due to the promise of AI-powered chatbot interaction, as illustrated in a recent study from the general HCI community (Wu et al., 2022). Previous focus on human interaction with buttons and displays and the user experience of those interactions, may be partly or in some cases wholly replaced by human-GenAI collaboration. CPS with HITL could thus in the future be studied

as human-CPS collaboration, in addition to applying traditional human-centered approaches of studying the usability and UX of the CPS with HITL.

Given that our approach was a literature review with an additional validation done by qualitative research interviews with industry experts, there are possibilities for future research to apply other methods. Firstly, a significantly larger validation study engaging a diverse range of industry experts from a variety of industries will perhaps add important nuances to the identification of gaps between the literature and industry experts. Second, the use of AI for literature selection and extraction is rapidly increasing (Wagner et al., 2022), which potentially can enable future studies to increase the range of the review to include more relevant social science and humanities approaches in the study of CPS with HITL.

6.4. Limitations

There are several limitations of this research, which we have tried to overcome. First, the literature could have been even broader by not including CPS with HITL as keywords, but we found that would have scoped the paper out of interest for CPS researchers. Second, the literature did not discuss how CPS systems accommodate vulnerable groups including individuals with physical or cognitive disabilities; however, we did identify issues with cognitive limitations in the literature reviewed, see Section 4.2.1. Third, the interviewed industry experts had technical backgrounds, but we included one with business background to get that perspective also represented in the viewpoints. Fourth, they came from tech providers, not end-user companies/clients so discussions of ST literature were somewhat limited from that position, which we tried to overcome by additional translation and interpretation during the interviews and during data analysis. Overall, we believe that the selection of literature and industry experts were valuable and helped produce an interesting analysis.

7. Conclusion

The answers provided by this systematic literature review of ST perspectives on CPS with HITL are that four perspectives: socio-technical capacities, a human-centered approach, human-computer interaction technologies, and human work design, are worthwhile including in CPS research on the environment, infrastructure, coordination, and decision making. In addition, we validated the literature review with interviews with industry experts, which revealed agreements and disagreements between their view and the ST perspectives on CPS with HITL. Future industry-oriented research may be better at theorizing CPS with HITL by integrating these perspectives in new control models. Among possible topics for future research are positioning of humans and technical devices in the CPS with HITL networks, bring in the rich variety of ideas in ST literature on how to enhance HITL, embrace a ST terminology, and do detailed field studies of manufacturing workers and their interactions with a full range of humans and technical devices in the CPS networks.

CRedit authorship contribution statement

Torkil Clemmensen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mahyar Tourchi Moghaddam:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Jacob Nørbjerg:** Writing – review & editing, Validation, Methodology.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Bing copilot in order to organize and edit the list of recommendations in the discussion. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data is available at Clemmensen, T., 2025. Cyber-physical systems with human-in-the-loop: A systematic review of socio-technical perspectives DATA. Zenodo, <http://dx.doi.org/10.5281/zenodo.14626314>.

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